Bandwidth Efficient Partial Authorized PSI

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***Abstract*—Recent attacks on private set intersection (PSI) and PSI-like protocols have demonstrated that input privacy can be compromised when parties maliciously choose their inputs—even in protocols proven secure against malicious adversaries. To counter such attacks, Authorized PSI (APSI) introduces a judge who authorizes the elements of the parties before the intersection is computed. Falzon and Markatou (PETS 2025) proposed Partial-APSI, a privacy-preserving variant of APSI that prevents revealing the entire set to a judge. Their Partial-APSI protocol requires significant bandwidth overhead due to the use of bilinear pairings and because the judge must sign each element in the input set. In this work, we present a bandwidth-efficient Partial-APSI protocol that outperforms Falzon and Markatou, both asymptotically and empirically. For example, for sets of size 2^{20}, we require around $21\times$ less bandwidth and are about $6\times$ faster over a LAN network. In addition to our protocol, we model the real-world behavior of rational parties through a game-theoretic analysis. We introduce payout mechanisms for detected cheating and establish lower bounds on their values, ensuring that the best strategy for rational parties is to provide honest input.**

1. Introduction

Private set intersection (PSI) is a secure multiparty computation technique that enables two or more parties to securely compute the intersection of their input sets while keeping the remainder of their inputs private. PSI has several real-world applications, including genome comparison [4, 55], profile matching [46], bot net detection [51], and private mobile contact discovery [10, 35, 40].

Another important use case that has already seen early-stage deployment is *ad conversion* [37, 38, 45]. A store and an ad provider can use PSI to privately compute the intersection of users who both viewed an ad and made a purchase. This enables the store to evaluate the ad's effectiveness without revealing sensitive customer data.

Numerous efficient PSI protocols have been proposed under various threat models. Some protocols are proven secure against semi-honest adversaries, where all parties are assumed to follow the protocol honestly but may attempt to learn additional information from the protocol transcript [36, 48, 53]. In contrast, other protocols have been proven secure against malicious adversaries, which may arbitrarily deviate from the protocol to compromise security [9, 54]. However, protocols proven secure in either threat model suffer from a key limitation: they *do not prevent* a party from maliciously selecting their input.

Several recent attacks against PSI(-like) protocols [22, 33, 39] demonstrate that when an adversarial party is allowed to choose their input arbitrarily, they can break input privacy. Additionally, many Bloom filter-based PSI protocols allow an adversarial party to infer membership of elements of the others' set not in the intersection [60].

One mitigation against such attacks is to introduce a trusted third-party judge to authorize the parties' inputs before computing the intersection (e.g., [7, 15]). This PSI variant, called *Authorized Private Set Intersection* (APSI), reveals the entire set to the judge. Recently, Falzon and Markatou [21] introduced *Partial Authorized Private Set Intersection* (Partial-APSI or PAPSI), where each party commits to their input set and reveals only a random subset selected by the judge for verification.

This selective disclosure helps preserve privacy while enabling input validation. As a result, Partial-APSI is especially well-suited for complying with regulatory frameworks that mandate data minimization. For example, according to Article 5(1)(c) of the European Union's General Data Protection Regulation (GDPR), organizations must ensure that only the minimum amount of personal data necessary for a specific purpose is collected, processed, and retained [24]. Data minimization is also a core tenet of the California Consumer Privacy Act (CCPA) [17]. Partial-APSI has several further applications, including:

Financial Auditing. Returning to the ad conversion example, we wish to ensure that the input set of the store contains only legitimate customers. To achieve this, a financial auditor can act as a judge and sample a subset of the customer list to verify its validity before the intersection. This approach is called *statistical audit sampling* and is a common auditing technique. PAPSI enables this process by only allowing the auditor access to the selected records, ensuring that the other customers' records remain hidden and eliminating unnecessary data sharing.

Contact matching. When a new user subscribes to a chat provider, they want to know which of their contacts use the application. The new user and the chat provider can use PSI to determine this intersection. In Authorized PSI, a telecom company can act as a judge. To verify the users of the chat provider, they can check if the user signed the terms of service.

Genomic Matching. Genome matching has become increasingly important for applications ranging from ancestry testing to blood transfusions. PSI offers a promising solution for genome matching. Traditionally, serological antibody-based methods have been used to match blood donors with recipients. However, complications can arise for patients who have antigens for which there are no

TABLE 1. COMPARISON WITH CLOSELY RELATED WORK. HERE, n IS THE SIZE OF THE RECEIVER'S SET, m IS THE SIZE OF THE SENDER'S SET, p IS THE FRACTION OF ELEMENTS REVEALED TO THE JUDGE, AND $k < n$ IS THE BRANCHING FACTOR OF THE VERKLE TREE.

Protocol		Security	Assumptions	Authorize		Intersect	
Citation	(P)APSI			Rounds	Communication	Rounds	Communication
DT10 [14]	APSI	Input mal.	ROM, RSA	2	$2n+2$	2	$2+2n+m$
DKT10 [13]	APSI	Malicious	ROM, RSA, DDH	2	$2n+2$	2	$1+5n+m$
DD15 [16]	APSI	Malicious	QR	2	$2n+2$	2	$2n+m$
FM25 [21]	APSI	Mal. receiver	DBDH	2	$2n+2$	1	$1+m$
FM25 [21]	PAPSI	Input mal.	ROM, DBDH, DDH	4	$2(p+1)n+2$	3	$1+3m$
Ours	PAPSI	Input mal.	ROM + DDH				
-Pointproof-Verkle tree			+ AGM, t-wBDH	4	$3+pn(2+\log_k(n))$	2	$1+2n+m$
-Merkle tree			-	4	$3+pn(2+\log_2(n))$	2	$1+2n+m$

serologic reagents. Recently, DNA-based genotyping has shown significant promise as a life-saving technique in reducing complications [61]. Partial-APSI can be valuable to match a blood donor and recipient from different hospitals. Hospitals trust a blood bank as a judge, but they also want to protect their patients' privacy. Partial-APSI allows the hospital to reveal a small percentage of patients.

The protocol proposed by Falzon and Markatou has a limitation: it has a high bandwidth usage. While the improvement of only partially revealing elements is promising, the high bandwidth costs are prohibative, thus raising the following question:

Can we design a bandwidth-efficient Partial-APSI protocol with sub-linear communication in the authorisation phase?

1.1. Contributions

We propose a Partial-APSI protocol that uses minimal bandwidth and significantly outperforms Falzon and Markatou [21]. For sets of size 2^{20}, we require approximately $21\times$ less bandwidth than [21]; furthermore, we reduce the runtime over a LAN network $6\times$.

We prove our protocol secure against a semi-honest sender and judge (i.e., they follow the protocol honestly but attempt to learn as much information as possible about the receiver's set) and an input-malicious or *augmented semi-honest* receiver (i.e., the receiver can maliciously select their input before both the authorization and intersection phases, but otherwise follows the protocol).

Traditional PSI threat models assume semi-honest or malicious adversaries, failing to capture the real-world nuances of adversaries seeking to maximize their utility. To this end, we apply the game-theoretic Adversarial Level Agreements (ALAs) framework of George and Kamara [25] to analyze how a rational receiver, acting in their own best interest, would behave when participating in our Partial-APSI protocol. We introduce a game-theoretic analysis examining how incorporating payouts for detected malicious inputs can incentivize a rational receiver to adhere to the protocol.

Our contributions can be summarized as follows:

- We propose a **bandwidth-efficient Partial-APSI** protocol. (Section 4)
- We support our protocol with **formal proofs of correctness and security**. (Sections 4.2 and 4.3)
- We provide an **open-source** Rust implementation of our protocol and experimentally **evaluate** it, showing we outperform prior work. (Section 6)
- We give a **game-theoretic analysis** of Partial-APSI and prove **lower bounds on the payouts** that incentivize a rational receiver to adhere to the protocol. (Section 5)
- We additionally prove that the Verkle tree commitment scheme is **hiding** and **binding**, which could be of independent interest (Section 3.3).

1.2. Related work

PSI has been extended in numerous ways to incorporate authorization mechanisms, leading to what is now commonly known as Authorized PSI (APSI). A foundational example of such an extension is the *PSI for certified* sets proposed by Camenisch and Zaverucha [7], where a trusted entity validates the participants' input sets. Concurrently, De Cristofaro et al. [15] explored a more targeted use case, *privacy-preserving policy-based information transfer* (PPIT), which enables receivers to retrieve specific items held by the sender based on third-party authorizations.

Authorized PSI (APSI) was formally introduced in [14] along with a linear-cost protocol that is secure against semi-honest adversaries. This work was later extended by De Cristofaro et al. [13], who proposed a maliciously secure scheme that also retained linear communication complexity. Kerschbaum [43] proposed an APSI protocol based on Bloom filters and homomorphic encryption. This was further optimized by Debnath et al. [16], who proposed protocols for both standard APSI and *APSI cardinality*, i.e., only the intersection size is revealed. Faber et al. [18] introduced a variant of APSI in which both parties must authorize their inputs and learn the intersection.

Policy-enhanced PSI (PPSI) was introduced by Stefanov et al. [56] to enable more complex authorization policies. Their work is secure against malicious adversaries, provided that the underlying PSI scheme used in a black-box manner is maliciously secure. Nagy et al. [52] proposed *Common Friends*, a social network application where users can determine if they are friends or share mutual friends. Their construction enforces access control in a privacy-preserving way and uses Bloom filters for efficiency. D'Arco et al. [12] proved that unconditionally secure size-hiding PSI is only possible in the APSI setting.

Sun et al. [57] gave a maliciously-secure PSI protocol that allows the sender to publish a one-time, linear-size encoding of its set and use it to compute intersections with multiple receivers. Ghosh et al. [26] proposed *private certifier intersection*, which allows multiple parties with certificates to discover shared certifiers and verify those, without revealing any information about certifiers that are not part of the intersection.

Recently, Falzon and Markatou [21] introduced *Partial-APSI* and presented one APSI protocol and one Partial-APSI. Another recent work by Goel et al. [27] introduced *Private Intersection over Committed (and Reusable) Sets*, which uses commitments to ensure that parties use the same input set across multiple runs. While this ensures consistency of the input across multiple runs, it does not guarantee that the parties use an honest input.

Yang et al. [62] propose an authorization phase, where none of the plaintext elements of the client are revealed. However, they require an additional party, an authority, as well as the judge. Yang et al. [62] do not provide an intersection phase.

Table 1 compares our protocol with closely related (P)APSI protocols that also comprise one judge and two parties. Compared to the authorization phase of [21] we use less bandwidth if $p < \frac{n \log_k(n)-1}{2n-1}$, since typically $2 \leq \log_k(n)$ (see also Fig. 8).

2. Preliminaries

2.1. Problem Definition & Threat Model

Partial-APSI consists of two phases. (i) **Authorization**: a Judge $\mathcal{J}$ requests a random fraction $p \in (0, 1]$ of the set of the receiver $\mathcal{C}$ and **authorizes** it. (ii) **Intersection:** the Sender $\mathcal{S}$ verifies that $\mathcal{C}$'s elements are authorized and then $\mathcal{C}$ and $\mathcal{S}$ compute the **intersection**. We consider one-sided PSI, where only $\mathcal{C}$ learns the intersection.

- The receiver $\mathcal{C}$ is an input-malicious (or augmented semi-honest) party. $\mathcal{C}$ does not deviate from the protocol, apart from potentially modifying their input at the beginning of the authorize and intersect phases.
- The Judge $\mathcal{J}$ is modeled as a semi-honest party. They do not deviate from the protocol, but may try to extract information about the non-revealed elements of $\mathcal{C}$.
- The sender $\mathcal{S}$ is modeled as a semi-honest party. They do not deviate from the protocol, but may also try to passively extract information about the elements of $\mathcal{C}$ from the protocol transcripts.

We prove security using the real-ideal paradigm. In the real world, the parties execute our protocol. In the ideal world, a trusted party executes the ideal functionality (Fig. 1) on inputs chosen by the receiver and the sender.

Definition 2.1. *Let* $\mathsf{View}^{\Pi}_{\mathcal{C}}(\mathcal{X}, \mathcal{Y})$, $\mathsf{View}^{\Pi}_{\mathcal{J}}(\mathcal{X}, \mathcal{Y})$, *and* $\mathsf{View}^{\Pi}_{\mathcal{S}}(\mathcal{X}, \mathcal{Y})$ *be the views of* $\mathcal{C}$, $\mathcal{J}$ *and* $\mathcal{S}$ *in the protocol* Π, *respectively. Let* $\mathsf{Out}^{\Pi}(\mathcal{X}, \mathcal{Y})$ *be the output of* Π *(i.e., the intersection) and* $O = f(\mathcal{X}, \mathcal{Y}, \mathcal{R})$ *be the output of the ideal functionality of Fig. 1. The protocol* Π *is* ***semi-honest secure for the*** $\mathcal{S}$ ***and*** $\mathcal{J}$ *and* ***augmented semi-honest secure for the*** $\mathcal{C}$ *if there exists PPT simulators* $\mathsf{Sim}_{\mathcal{C}}$, $\mathsf{Sim}_{\mathcal{S}}$ *and* $\mathsf{Sim}_{\mathcal{J}}$ *with input* 1^{λ}, *such that for all inputs* $\mathcal{X}, \mathcal{Y}$, *for all public parameters* $\mathcal{R}$ *and* p, *and for all index sets* $\mathcal{I} \subseteq [n]$ *such that* $|\mathcal{I}| = pn$:

$$\left(\mathsf{View}^{\Pi}_{\mathcal{J}}(\mathcal{X}, \mathcal{Y}), \mathsf{Out}^{\Pi}(\mathcal{X}, \mathcal{Y})\right) \overset{c}{\approx} (\mathsf{Sim}_{\mathcal{J}}(\{x_i\}_{i \in \mathcal{I}}, \mathcal{I}, |\mathcal{X}|), O)$$

$$\left(\mathsf{View}^{\Pi}_{\mathcal{S}}(\mathcal{X}, \mathcal{Y}), \mathsf{Out}^{\Pi}(\mathcal{X}, \mathcal{Y})\right) \overset{c}{\approx} (\mathsf{Sim}_{\mathcal{S}}(\mathcal{Y}, |\mathcal{X}|), O)$$

$$\mathsf{View}^{\Pi}_{\mathcal{C}}(\mathcal{X}, \mathcal{Y}) \overset{c}{\approx} \mathsf{Sim}_{\mathcal{C}}(\mathcal{X}, |\mathcal{Y}|, O)$$

Ideal Functionality for Partial-APSI

Parties and inputs:
Receiver $\mathcal{C}$ inputs sets $\mathcal{X}$ for Authorize and $\mathcal{X}'$ for Intersect, sender $\mathcal{S}$ inputs set $\mathcal{Y}$; judge $\mathcal{J}$ inputs $\perp$.

Parameters:
Set sizes $|\mathcal{X}|$, $|\mathcal{Y}|$, probability $p \in (0, 1]$, regulation knowledge $\mathcal{R}$

Functionality:
$\mathbf{T}$ receives an Authorize request of $\mathcal{X}$ from $\mathcal{C}$:

- $\mathbf{T}$ samples $\mathcal{I} \subseteq_{\$} [n]$ such that $|\mathcal{I}| = \lceil pn \rceil$ where $n = |X|$, and then either accepts or rejects the elements in $\{x_i \mid x_i \in \mathcal{X}\}_{i \in \mathcal{I}}$ based on $\mathcal{R}$.
- If the elements are consistent with $\mathcal{R}$, $\mathbf{T}$ replies `Accept` to $\mathcal{C}$ and remembers that $\mathcal{X}$ is authorized. Otherwise, $\mathbf{T}$ replies `Reject` to $\mathcal{C}$.

$\mathbf{T}$ receives a request from party $\mathcal{C}$ with input $\mathcal{X}'$ to perform set Intersection with the sender $\mathcal{S}$.

- If $\mathcal{X}' = \mathcal{X}$, $\mathbf{T}$ forwards the request to $\mathcal{S}$.
- $\mathcal{S}$ sends a set $\mathcal{Y}$ to $\mathbf{T}$.
- $\mathbf{T}$ computes the intersection $\mathcal{X} \cap \mathcal{Y}$ and sends this and the set size $|\mathcal{Y}|$ to $\mathcal{C}$.

Figure 1. Ideal Functionality for Partial-APSI, $\mathcal{F}_{\mathsf{Partial\text{-}APSI}}$.

2.2. Computational assumptions

Our protocol's security depends on the Decisional Diffie-Hellman assumption in the random oracle model (ROM). We use vector commitments as a black-box. Additional assumptions can be found in Appendix B.

Definition 2.2. Decisional Diffie-Hellman (DDH) assumption [5]*: Given a generator* g *of group* $\mathbb{G}$ *of prime order,* $a, b, c \in \mathbb{Z}$ *and* $c \neq ab$. *It holds that for all probabilistic polynomial time algorithms* $\mathcal{A}$ *and for security parameter* $\mathsf{negl}(\lambda)$

$$\Big|\Pr[\mathcal{A}(g, g^a, g^b, g^{ab}) = \text{true}] - \Pr[\mathcal{A}(g, g^a, g^b, g^c) = \text{true}]\Big| \leq \mathsf{negl}(\lambda).$$

2.3. Signature Scheme

Authorized elements will be marked with a signature.

Definition 2.3. *A* digital signature scheme *consists of three algorithms* $(\mathsf{KeyGen}, \mathsf{Sign}, \mathsf{Verify})$*:*

- $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^\lambda)$ *takes as input a security parameter* λ *and outputs a public-private key pair* $(\mathsf{pk}, \mathsf{sk})$.
- $\sigma \leftarrow \mathsf{Sign}(\mathsf{sk}, m)$ *takes as input a secret key* sk *and message* m*, and outputs a signature* σ.
- $b \leftarrow \mathsf{Verify}(\mathsf{pk}, \sigma, m)$ *takes as input a signature* σ*, a message* m*, and public key* pk *and outputs a bit* $b \in \{1, 0\}$.

A digital signature scheme is correct if it holds for any message m that $\mathsf{Verify}(\mathsf{pk}, \mathsf{Sign}(\mathsf{sk}, m), m) = 1$. We require the digital signature scheme to be existential unforgeable against chosen message attacks (EUF-CMA), which means that an adversary has at most negligible probability of forging a valid signature on a message that they have not seen before.

3. Vector commitment schemes

Commitment schemes are designed so that a party (the prover) can commit to a secret value. Later, the prover can reveal the secret value and prove that the commitment belongs to this value. It is also required that the commitment is binding; the prover cannot provide a different value with the same commitment. Additionally, a commitment can be hiding, i.e., the verifier should not be able to distinguish commitments of two different values. As an extension of commitment schemes, Catalano and Fiore [8] introduced vector commitment schemes (VCS) that allow for committing to a vector of values.

Definition 3.1. *A vector commitment scheme is a tuple of algorithms* $(\mathsf{Setup}, \mathsf{Commit}, \mathsf{Open}, \mathsf{V})_{vcs}$

- $\mathsf{pp}_{vcs} \leftarrow \mathsf{Setup}_{vcs}(k)$ *This algorithm outputs the public parameters supporting* k *messages.*
- $\mathsf{C}, \mathsf{aux} \leftarrow \mathsf{Commit}_{vcs}(m_1, .., m_k, \mathsf{pp}_{vcs})$ *On a sequence of* k *messages* $m_1 ... m_k$ *and the public parameters* pp_{vcs}*, this algorithm outputs a commitment string* C *and auxiliary information*
- $\Lambda_i \leftarrow \mathsf{Open}_{vcs}(m, i, \mathsf{aux})$ *On input of message* m*, index* i *and auxiliary information, the open algorithm returns a proof* Λ_i.
- $\{0, 1\} \leftarrow \mathsf{V}_{vcs}(\mathsf{C}, m, i, \Lambda_i)$ *This algorithm outputs 1 if and only if* Λ_i *is a valid proof that* C *was created with message* m *on the* i*'th index.*

A vector commitment scheme is correct *if*

$$\Pr\left[\begin{array}{c}\mathsf{V}_{vcs}(\mathsf{Commit}_{vcs}(m_1, .., m_k, \mathsf{pp}_{vcs}),\\ m_i, i, \mathsf{Open}_{vcs}(m_i, i, \mathsf{aux}) = 1)\end{array}\right] \geq 1 - \mathsf{negl}(\lambda).$$

We formally define *position-binding* and *hiding* below.

Definition 3.2. *A vector scheme VCS* $= (\mathsf{Setup}, \mathsf{Commit}, \mathsf{Open}, \mathsf{V})_{vcs}$ *is* position binding *if for every PPT adversary* $\mathcal{A}$*,* $(\mathsf{C}, i, m, m', \Lambda, \Lambda') \leftarrow \mathcal{A}$ *and* $m \neq m'$ *it holds*

$$\Pr\left[\begin{array}{c}\mathsf{V}_{vcs}(\mathsf{C}, m, i, \Lambda) = 1\\ \wedge\\ \mathsf{V}_{vcs}(\mathsf{C}, m', i, \Lambda') = 1\end{array}\right] \leq \mathsf{negl}(\lambda).$$

Definition 3.3. *A vector scheme VCS* $= (\mathsf{Setup}, \mathsf{Commit}, \mathsf{Open}, \mathsf{V})_{vcs}$ *is* hiding *if for every PPT adversary* $\mathcal{A}$ *and any two sequences of messages* $V_1 = [m_1, ..., m_k]$ *and* $V_2 = [m'_1, ..., m'_k]$ *it holds that*

$$\Pr\left[\begin{array}{c|l} b = b' & \begin{array}{l}\mathsf{pp}_{vcs} \leftarrow \mathsf{Setup}_{vcs}(k)\\ (V_1, V_2) \leftarrow \mathcal{A}\\ b \leftarrow\!\!\$\ \{0, 1\}\\ \mathsf{C} \leftarrow \mathsf{Commit}_{vcs}(V_i, \mathsf{pp}_{vcs})\\ b' \leftarrow \mathcal{A}(\mathsf{C})\end{array}\end{array}\right] \leq \mathsf{negl}(\lambda),$$

even after opening one or more positions.

We additionally require the vector commitment scheme to be *concise*.

Definition 3.4. *The vector scheme VCS is* concise *if the output sizes of* Commit_{vcs} *and* Open_{vcs} *are constant and independent of the length of the vector.*

In the following subsections, we give examples of (concise) vector commitment schemes.

3.1. Merkle tree

In 1989, Merkle [49] proposed a digital signature using a complete binary tree. The tree is position binding and can thus be used as a vector commitment scheme. The prover constructs the tree from bottom to top. Each leaf node i contains the value x_i. Then the prover hashes each pair of children to form the parent nodes. This is iterated until the root. The commitment of the vector is the value of the root. If the verifier queries an index i, the prover sends the value x_i and the sibling node, along with all hash values of the sibling nodes on the path to the root. To verify, the verifier starts by hashing the two leaf values. Then, the verifier iteratively hashes the new value together with the value of the sibling node. The iteration ends at the root. The verifier finally compares the last hash to the commitment; if they are equal, the verifier approves the proof. The Merkle tree is binding if the hash functions are modeled as a random oracle [49]. One downside is that the tree is not hiding, as the sibling node of the leaf node is revealed. One can solve this by hashing the value to form the leaves. The second disadvantage is that the proof of one element is logarithmic in the size of the vector, and hence, this commitment scheme is not concise.

3.2. Pointproof vector commitment

Gorbunov et al. [31] proposed a concise pairing-based scheme based on the vector commitment of Libert and Yung [47]. Let $(\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T)$ be a bilinear group and let g_1 be a generator of $\mathbb{G}_1$. To commit to a vector $V = [m_1, \ldots, m_k]$, we construct a polynomial where the coefficient of x^i is m_i for each $i \in [k]$. The commitment to V is given by g_1 raised to the value of this polynomial evaluated at a secret point. The scheme is binding in the algebraic group model, random oracle model, and under the t-weak bilinear Diffie-Hellman assumption (for definitions, see Appendix B). Proofs of binding and hiding can be found in [31].

3.3. Verkle tree commitment scheme

A Verkle tree [44] is similar to a Merkle tree; the difference is that the parent node is a vector commitment

to its children. Furthermore, the Verkle tree does not have to be a binary tree, and any branching number k can be chosen. This choice determines the proof size and computation time. To the best of our knowledge, no formal proof exists of the security of a Verkle tree, and we thus provide the proof in this section.

To construct a Verkle tree, one first chooses a concise vector commitment scheme and a branching factor k. Then one takes the messages m and hashes them to the field over which the VCS is defined. The parent node of k successive (leaf) nodes is defined as the single token commitment of the VCS commitment of these nodes. An example is pictured in Fig. 2. In this example, $n = 9$, the width k is 3, and the messages m_i are hashed to the leaf nodes $\hat{x}_i$. In the next layer, C_1 is the vector commitment of $[\hat{x}_1, \hat{x}_2, \hat{x}_3]$ similary for C_2 and C_3, finally the root is a commitment of $\mathsf{C}_1, \mathsf{C}_2, \mathsf{C}_3$. To prove the presence of m_7 in the tree, the prover provides a proof $\lambda_1 = \mathsf{Open}_{vcs}(\hat{x}_7, 1)$, and $\lambda_2 = \mathsf{Open}_{vcs}(\mathsf{C}_3, 3)$. The verifier verifies $\mathsf{V}_{vcs}(\mathsf{C}_3, \hat{x}_7, 1, \lambda_1)$ and $\mathsf{V}_{vcs}(\mathsf{R}, \mathsf{C}_3, 3, \lambda_2)$. The proof path is blue in Fig. 2.

A Verkle tree commitment scheme is a tuple of algorithms $(\mathsf{Setup}, \mathsf{Commit}, \mathsf{Open}, \mathsf{V})$ defined as:

- $(\mathsf{pp}_{vcs}, k) \leftarrow \mathsf{Setup}_{tree}(k)$ On input k, the width of the tree, this algorithm outputs the public parameters pp_{vcs} of the underlying vector commitment scheme and the width of the tree.
- $\mathsf{R} \leftarrow \mathsf{Commit}_{tree}(m_1, .., m_n, \mathsf{pp})$ On a vector of n messages this algorithm calculates Commit_{vcs} of the underlying vector commitment scheme for each k sequential leaves. Then the algorithm calculates Commit_{vcs} for k sequential parent nodes. This tree structure is repeated until the root. The output of this algorithm is the root R.
- $\Lambda_i \leftarrow \mathsf{Open}_{tree}(m, i, \mathsf{aux}_{vcs})$ The algorithm initializes an empty list Λ_i. On input of a message and an index, this algorithm performs iteratively $\Lambda_j \leftarrow \mathsf{Open}_{vcs}$ for each parent-child pair along the treepath of the requested leaf i. For each opening on commitment node C with child node c on index j of the vector commitment, the algorithm adds the tuple $(\mathsf{C}, c, j, \Lambda_j)$ to Λ_i. The output is Λ_i.
- $\{0, 1\} \leftarrow \mathsf{V}_{tree}(\mathsf{R}, m, i, \Lambda_i)$. On input of a message, an index, a proof, and the root, the algorithm performs V_{vcs} for each parent-child pair along the tree path. The algorithm outputs 1 if and only if all proofs are correct and the indices follow the path in the tree.

Remark. Note that to ensure security, a fully populated Verkle tree is required. To solve this, either dummy messages need to be added. Alternatively, a forest of trees could be used, where the n messages are split across several filled trees.

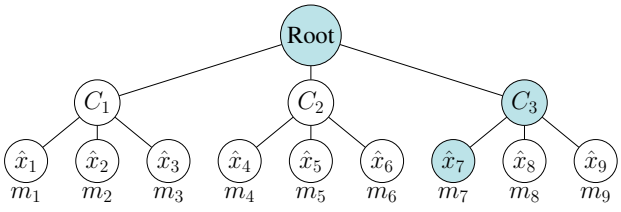


Figure 2. Verkle tree example with branching factor 3.

Unfortunately, this vector commitment scheme is not concise as the proof is logarithmic in n, ie. $O(\log_k(n))$.

Lemma 3.5. *The Verkle tree is* position binding *if and only if the underlying vector commitment scheme is.*

Proof. Assume towards a contradiction that the Verkle tree is not position binding. There exists an adversary $\mathcal{A}$ who, in polynomial time, produces two valid openings Λ, Λ' to two different messages at the same position:

$$(\mathsf{R}, i, m, m', \Lambda, \Lambda') \leftarrow \mathcal{A}.$$

We can then construct an adversary $\mathcal{B}$ that breaks the position binding for the VCS. $\mathcal{B}$ selects the width of the verkle tree, k, and samples public parameters for the vector commitment scheme accordingly. $\mathcal{B}$ forwards the public parameters to adversary $\mathcal{A}$. The adversary can break the position binding property of the verkle tree, and $\mathcal{A}$ returns $(\mathsf{R}, i, m, m', \Lambda, \Lambda')$ for some $i \leq n$. Λ and Λ' are lists of proofs of commitments along the tree paths, and $\mathcal{B}$ checks the proofs one by one. The first proof λ_i in Λ is a tuple of the form $(\mathsf{R}, c, i_1, \Lambda_c)$ the first tuple of Λ' is $(\mathsf{R}, c', i_1, \Lambda'_{c'})$. Note that the values of R and i are the same, because the tree path is unique. If the values c and c' or the values Λ_c and $\Lambda'_{c'}$ differ, then $\mathcal{B}$ outputs the tuple $(\mathsf{R}, i_1, c, c', \Lambda_c, \Lambda'_{c'})$. If, on the contrary, both tuples are equal, then $\mathcal{B}$ continues its search from the root to the leaf. At least one of the proofs should be different, and thus $\mathcal{B}$ breaks the position binding of the VCS. The adversary $\mathcal{B}$ works in polynomial time as the tree path is logarithmic to the input, and $\mathcal{A}$ works in polynomial time. □

Lemma 3.6. *The Verkle tree is* hiding *if and only if the underlying vector commitment scheme is.*

This lemma can be proven similarly to Lemma 3.5.

4. Partial-APSI Protocol

We now present our partially authorized private set intersection scheme. Partial authorization requires the Receiver to commit to the set $\mathcal{X}$ and send the commitment to the judge. The judge then challenges the Receiver to reveal a $p \in (0, 1]$-fraction subset of its input. The previous Partial-APSI scheme [21] results in a significant bandwidth overhead in the authorization phase, since the commitment and signatures are linear in $|\mathcal{X}|$. To optimize, we use a concise commitment scheme, resulting in a single token for both the commitment and signature.

4.1. Protocol overview

Our protocol's public parameters are: a cryptographically secure hash function H needed to blind the elements of the Receiver and sender, and the public parameters of the vector commitment scheme pp_{vcs}. Before the protocol, the judge generates a secret key sk and public key pk for the signature. The receiver has input $\mathcal{X}$ of size n and the sender has input $\mathcal{Y}$ of size m, their elements are from $\{0, 1\}^*$. Whether an element of the Receiver is authorized depends on public regulation knowledge $\mathcal{R}$.

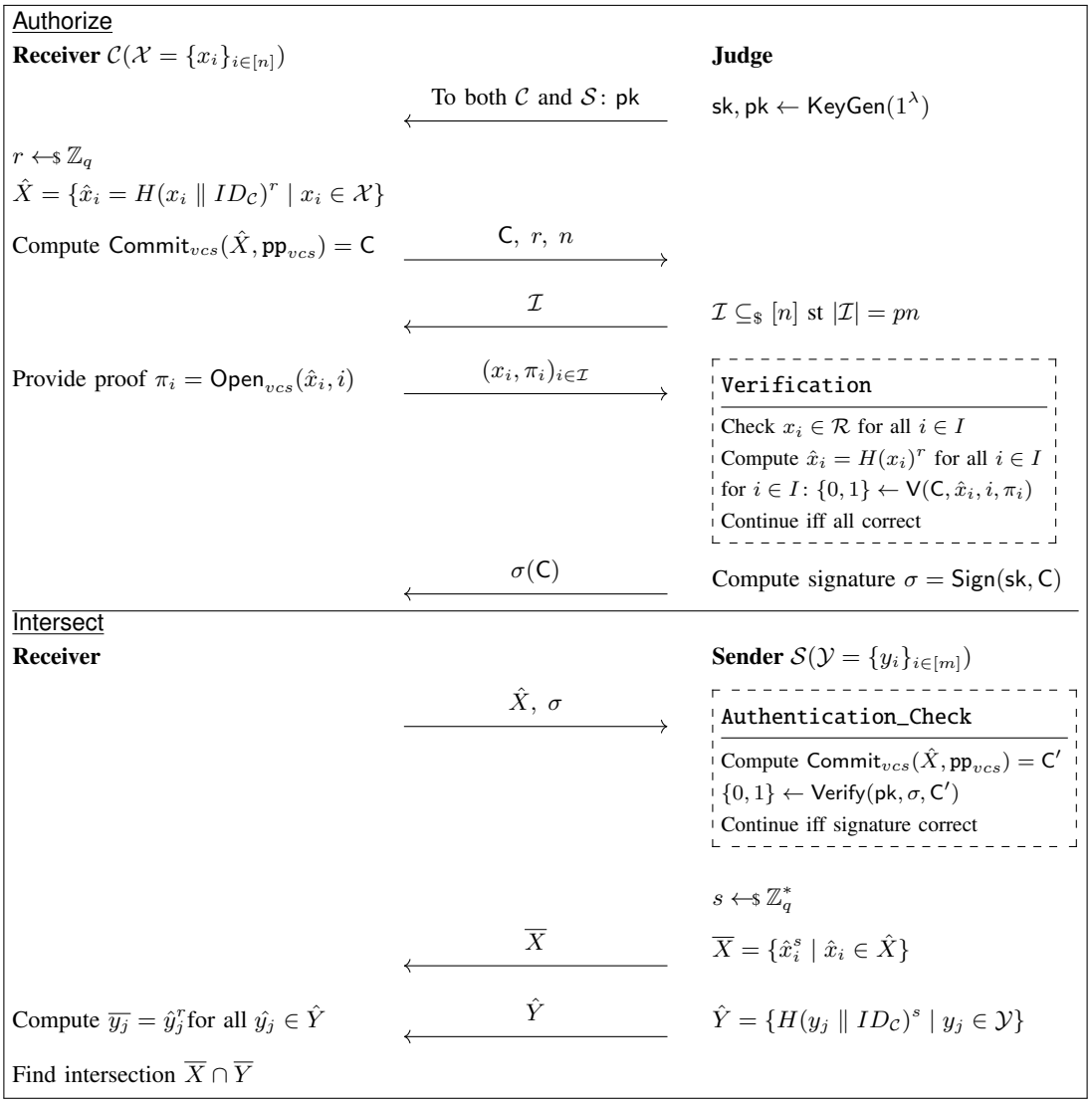


Figure 3. Partially Authorized PSI, with as public parameters: prime number q, hash function $H\colon \{0,1\}^* \to \mathbb{Z}_q^*$, public regulation knowledge $\mathcal{R}$. Moreover, the used vector commitment scheme VCS together with pp_{vcs} should be public.

Here we give a high-level overview of the protocol; a detailed version can be found in Fig. 3. Our protocol starts with the authorization phase:

1) The Receiver starts by blinding all of their elements ($\mathcal{X}$) and raises them to a random r.
2) Then, the Receiver commits the blinded values in the VCS, and produces a vector commitment C.
3) After the judge receives the commitment, C, they sample a challenge of indices $\mathcal{I}$ and forward it to the receiver.
4) The Receiver sends the requested items in plaintext and proves that they committed to these.
5) If the judge approves the elements and verifies the commitment proofs, the judge signs the commitment, C, and returns the signature to the Receiver.

After Authorize, the Receiver starts communicating with the sender to compute the intersection.

1) The Receiver sends the sender all blinded elements and the signed commitment.
2) The sender computes $\mathsf{C} = \mathsf{Commit}_{\mathsf{vcs}}$ with the blinded elements. Then they check the signature of C. This way, the sender ensures that the judge authorized this set.
3) If the sender approves the signature, the parties continue with a Diffie-Hellman-based PSI protocol. To this end, the sender samples a random s and exponentiates both sets to the power s. The Receiver raises the set of the sender to r and can compute the intersection.

The following lemma shows that the bandwidth of the authorization phase primarily depends on the proof sizes.

Lemma 4.1. *The receiver-judge communication is:*

$$O(pn|\mathsf{Open}_{vcs}|). \tag{4.1}$$

Proof. (1) The protocol starts with the Receiver who sends integers r, n and commitment $\mathsf{C} = \mathsf{Commit}_{\mathsf{vcs}}$. For a concise VCS, this is of constant size. (2) The judge samples pn indices and sends those to the receiver; this is of order pn. (3) The Receiver responds with the pn requested elements in plaintext and their proof of commitment, resulting in an overall complexity of:

$$O(|\mathsf{Commit}_{vcs}| + |r| + |n| + pn(2 + |\mathsf{Open}_{vcs}|))$$

Omitting the constants gives the lemma. □

For $p < \frac{n \log_k(n)-1}{2n-1}$, we achieve better authorization bandwidth than Falzon and Markatou [21].

Lemma 4.2. *The receiver-sender communication is of order*

$$O(2n + m) \tag{4.2}$$

Proof. (1) The Receiver sends the signature of the commitment, which is of a constant size. Further, the Receiver sends the blinded set $\hat{X}$ of order n. (2) The sender responds with the set $\hat{X}$ blinded again, and it own blined set $\hat{Y}$, of order $n + m$, this concludes the proof. □

4.2. Correctness of the protocol

Theorem 4.3. *The partial-APSI protocol in Fig. 3 outputs $\mathcal{X} \cap \mathcal{Y}$, where $\mathcal{X}$ is the Receiver $\mathcal{C}$'s input and $\mathcal{Y}$ is the sender $\mathcal{S}$'s input if H is a collision-resistant hash function.*

Proof. The intersection is correct because for any $x_i = y_j$, where $x_i \in \mathcal{X}$ and $y_j \in \mathcal{Y}$, we see the following:

$$\overline{x}_j = (H(x_i \parallel ID_{\mathcal{C}}))^{rs} = (H(y_j \parallel ID_{\mathcal{C}}))^{sr} = \overline{y}_j,$$

this indicates that any element x_i in the intersection $\mathcal{X} \cap \mathcal{Y}$, is indeed in the final intersection. For any element x_i not in the intersection, it happens only with negligible probability that there exists a y_j such that $(H(x_i \parallel ID_{\mathcal{C}}))^{rs} = (H(y_j \parallel ID_{\mathcal{C}}))^{sr}$, due to the collision-resistant hash function. □

4.3. Security of the protocol

Theorem 4.4. *The Partial-APSI protocol in Fig. 3 is secure against a semi-honest judge if the vector commitment scheme is hiding and H is a collision-resistant hash function.*

Proof. We prove this using a hybrid argument. Upon input of $1^\lambda, \{x_i\}_{i\in\mathcal{I}}, \mathcal{I}, |\mathcal{X}|, \mathcal{R}, p$, the simulator proceeds as follows:

- Sample element $r \leftarrow\$ \mathbb{Z}_q^*$.
- Define a set $\hat{X}$ of size $|\mathcal{X}|$ such that the i^{th} element is defined as $\hat{x}_i = H(x_i \| ID_{\mathcal{C}})^r$ if $i \in \mathcal{I}$ and $\hat{x}_i \leftarrow\$ \mathbb{Z}_q^*$ otherwise.
- Compute $\mathsf{C} = \mathsf{Commit}_{\mathsf{vcs}}$. Forward C, r, and n to the judge.
- Upon receiving $\mathcal{I}$, compute the commitment proof π_i of $\hat{x}_i$ and forward $(x_i, \pi_i)_{i\in\mathcal{I}}$ to the judge.

We now consider the following sequence of hybrids.
Hybrid 0**:** The real honest interaction.
Hybrid 1**:** The same as the previous Hybrid, except the output is replaced with the functionality output $f(\mathcal{X}, \mathcal{Y}, \mathcal{R})$. By correctness of the protocol (Theorem 4.3), this is computationally indistinguishable from Hybrid 0.
Hybrid 2**:** The same as the previous Hybrid, except sample $r \leftarrow\$ \mathbb{Z}_q^*$; since r is distributed the same as in the previous Hybrid, then the hybrids are indistinguishable.
Hybrid $(3, k)$ **for all** $k \le n - |\mathcal{I}|$**:** Let $j \in [n] \setminus \mathcal{I}$ be the smallest index in $\mathcal{X}$ such that $\hat{x}_j$ is computed as in the real world. This Hybrid is the same as the previous Hybrid, but $\hat{x}_j$ is replaced with a randomly sampled value $\hat{x}_j \notin \mathcal{X} \cap \mathcal{Y}$. This value changes the commitment, but since the VCS is hiding, the $(k-1)^{\text{th}}$ and k^{th} hybrids are computationally indistinguishable. Observe that the final Hybrid is identical to the simulator, and the proof follows. □

Theorem 4.5. *The Partial-APSI protocol in Fig. 3 is secure against a semi-honest sender if the hash function H is collision-resistant, the DDH assumption holds, and the vector commitment scheme is binding and hiding.*

Proof. We prove this using Hybrid arguments. Our simulator has input $1^\lambda, \mathcal{Y}, |\mathcal{X}|$ and works as follows:

- The simulator samples without replacement $|X|$ elements of $\mathbb{Z}_q^*$ to simulate $\hat{X}$.
- Compute the commitment $\mathsf{C} = \mathsf{Commit}_{\mathsf{vcs}}(\hat{X})$.
- The simulator generates the key pair $(\mathsf{sk}, \mathsf{pk}) \leftarrow \mathsf{KeyGen}(1^\lambda)$ and computes $\sigma(\mathsf{C}) \leftarrow \mathsf{Sign}(\mathsf{sk}, \mathsf{C})$.

Hybrid 0**:** The real interaction; the parties run the protocol honestly.
Hybrid 1**:** The same as the previous Hybrid, except we abort if there exists a pair $i, j \in [n]$, such that $\hat{x}_i = \hat{x}_j$. This means that there is a collision in the hash function, which happens with negligible probability because H is a collision-resistant hash function; thus, this Hybrid is indistinguishable from Hybrid 0.
Hybrid 2**:** The same as the previous hybrid, except that r is sampled from $\mathbb{Z}_q$. Since the sampling distribution is the same, the hybrids are indistinguishable from one another.
Hybrid $(3, k)$**:** Similar to the previous Hybrid, but $\hat{x}_k$ is replaced by a value $x' \leftarrow\$ \mathbb{Z}_q^*$, resampling as needed to avoid collisions within $\hat{X}$. Since $\mathbb{Z}_q^*$ is cyclic there exists a generator $g \in \mathbb{Z}_q^*$ and integer $a \in \mathbb{Z}_q$ such that for x_k we have $H(x_k \| ID_{\mathcal{C}}) = g^a$. Now, consider the tuple

$$(g, H(x_k \parallel ID_{\mathcal{C}}), g^r, H(x_k \parallel ID_{\mathcal{C}})^r) = (g, g^a, g^r, g^{ar})$$

and observe that, by the DDH assumption, this triple must be indistinguishable from a random group element g^b.

$$(g, H(x_k \parallel ID_{\mathcal{C}}), g^r, x') = (g, g^a, g^r, g^b).$$

Thus by the DDH assumption, $H(x_k \parallel ID_{\mathcal{C}})^r$ is indistinguishable from randomly sampled group element x'.
Moreover, the VCS is hiding, so no adversary can detect the replacement of values. Further note that since $\mathbb{Z}_q^* \gg |\hat{X}|$, resampling can be done in polynomial time, and thus this Hybrid runs in polynomial time and is indistinguishable from the previous Hybrid.
Hybrid $(2, n)$ is indistinguishable from our simulator. Since the simulator is indistinguishable from the real world, we conclude that the partial-APSI protocol is secure against a semi-honest sender. □

Theorem 4.6. *The partial-APSI protocol in Fig. 3 is secure against an input-malicious receiver $\mathcal{C}$ if the signature is EUF-CMA secure, the DDH assumption holds, the hash functions are modelled as a random oracle, and the vector commitment scheme is binding.*

Proof. We construct the simulator, which takes as input $1^\lambda, \mathcal{X}, |\mathcal{Y}|, \mathcal{R}, p$, and $O = f(\mathcal{X}, \mathcal{Y}, \mathcal{R})$, as follows:

- The simulator models the hash function H as a random oracle and stores the result in a table T. In particular, upon each query q to H, it checks

if a hash value h exists such that $h = T[q]$. If so, return h, otherwise, it samples a random h, return h and record $T[q] = h$.

- To simulate the judge:
  - The simulator generates a key pair $(\mathsf{sk}, \mathsf{pk}) \leftarrow \mathsf{KeyGen}(1^\lambda)$.
  - Upon an authorization request and receiving commitment C, the simulator samples a set $\mathcal{I}$ of size $|\mathcal{I}| = pn$ along a uniform distribution and forwards it to the Receiver.
  - Upon receiving the elements and proofs from the Receiver, the simulator checks the proofs and uses $\mathcal{R}$ to verify the elements.
  - If no malicious elements are detected and the proofs are correct, Sim computes the signature $\sigma(\mathsf{C}) \leftarrow \mathsf{Sign}(\mathsf{sk}, \mathsf{C})$. Otherwise, the simulator outputs `Reject`.
- To simulate the sender $\mathcal{S}$, on input $\hat{X}$ and $\sigma(\mathsf{C})$ from the receiver, the simulator does the following:
  - Compute C' with $\mathsf{Commit}_{\mathsf{vcs}}$ and $\hat{X}$. Use knowledge of C from authentcation to check if $\mathsf{C}' \stackrel{?}{=} \mathsf{C}$. Abort if they are not equal. Then verify the signature $\sigma(\mathsf{C})$ and abort if and only if the verification fails.
  - The simulator samples a random $s \in \mathbb{Z}_q$
  - Initalizes empty set $\hat{Y}$. For each $y \in \mathcal{X} \cap \mathcal{Y}$, make hash query $H(y \parallel ID_{\mathcal{C}})$, raises it to the power s, and adds it to $\hat{Y}$. It then pads $\hat{Y}$ up to size $|\mathcal{Y}|$ by sampling random elements from $\mathbb{Z}_q^*$ and shuffles $\hat{Y}$.
  - Initalizes empty set $\overline{X}$. For all $\hat{x} \in \hat{X}$, the simulator raises it to the power of s and adds it to $\overline{X}$. Forwards $\hat{Y}$ and $\overline{X}$ to the receiver.

We now consider the following series of hybrids:

Hybrid 0**:** The real honest interaction.

Hybrid 1**:** The same as the previous Hybrid, except the key pair is changed by $(\mathsf{sk}, \mathsf{pk}) \leftarrow \mathsf{KeyGen}(1^\lambda)$. The keys for the signature scheme are also generated along the same distribution, and thus the public key pk and the signature $\sigma(\mathsf{C})$ are indistinguishable from the real world.

Hybrid 2**:** The same as the previous hybrid, except $\mathcal{I}$ is sampled along a uniform distribution. It is generated along the same distribution as in the real world, and thus this Hybrid is indistinguishable from the previous one.

Hybrid 3**:** During this Hybrid, we check if any malicious elements are revealed and verify the commitment proofs. The probability of detecting a malicious element is $A = \binom{n-\mu n}{pn} / \binom{n}{np}$ (5.1), where μ is the fraction of malicious elements in the Receiver's input. This is the same probability for the Receiver being caught as in Hybrid 1. Note moreover that the Receiver is unable to cheat because the VCS is binding. The hybrids are thus indistinguishable.

Hybrid 4**:** The same as the previous Hybrid, except we compute commitment, C', and reject if $\mathsf{C} \neq \mathsf{C}'$ or the verification of $\sigma(\mathsf{C}')$ fails. We will find $\mathsf{C} \neq \mathsf{C}'$ with high probability if and only if the set $\hat{X}$ is not the same as the set $\hat{X}$ used during the authorization phase. If any changes are made, the binding property of the vector commitment scheme ensures that the commitment changes. If the Receiver changed the signature, the unforgability ensures that $\mathsf{Verify}(\mathsf{pk}, \sigma(\mathsf{C}), \mathsf{C})$ returns 0. We conclude that we approve if and only if the input set for $\mathsf{Intersect}$ was the same as for $\mathsf{Authorize}$. This Hybrid is therefore indistinguishable from the previous Hybrid.

Hybrid 5**:** During this hybrid game, s is sampled from $\mathbf{Z}_q$. It is generated along the same distribution; thus, this Hybrid is indistinguishable from the previous Hybrid.

Hybrid $(6, k)$ **for all** $k \leq m$**:** Same as the previous hybrid, but for all $y_k \notin \mathcal{X} \cap \mathcal{Y}$ a random value is sampled. Under the assumption that the hash function is modelled as a random oracle and the DDH assumption holds, this random value appears indistinguishable from $H(y \parallel ID_{\mathcal{C}})^s$ and the hybrids are indistinguishable.

Hybrid $(7, k)$ **for** $k \leq n$**:** The same as the previous Hybrid, except we abort if there is a collision, in particular if $H(y_k \| ID_{\mathcal{C}}) = H(y_j \| ID_{\mathcal{C}})$ for some $j \leq k$ or $H(y_k \| ID_{\mathcal{C}}) = H(x_i \| ID_{\mathcal{C}})$ for some $i \leq n$. This happens with probability $\frac{mn}{q}$ where $\mathbb{Z}_q^*$ is the field and m, n are the set sizes of $\mathcal{S}$ and $\mathcal{C}$. This is negligible in the group size and is indistinguishable from the previous Hybrid.

We observe that Hybrid $(7, n)$ is the same as the simulator, which concludes the proof. □

5. Game-Theoretic Evaluation

As noted in [2], modeling parties as semi-honest or malicious does not fully capture how adversaries may act in the real world; often, people are driven by self-gain or may try to cheat if they believe they will likely not be caught. Consider *rational actors* who act in their best interest and seek to maximize utility. Halpern and Teague [34] were the first to study rational adversaries in the context of secure multi-party computation, prompting a line of research (e.g., [3, 32] and [42] for an overview). Aumann and Lindell [2] proposed the concept of a *covert adversary* as an extension of the rational adversary, to capture adversaries who are willing to cheat as long as they are unlikely to be caught. This model sits between the semi-honest and malicious settings, providing a more realistic representation of adversarial behavior.

In this section, we analyze our PAPSI protocol through a game-theoretic lens. We apply the framework of George and Kamara [25] to model the behavior of a covert adversary. Their framework introduces *contracts* into protocols, which specify penalties that parties must pay if caught cheating. Covert adversaries act in their best interest, so the risk of penalties discourages them from deviating. George and Kamara show that such contracts can lead to favorable outcomes by incentivising honest behavior.

We investigate how to augment a Partial-APSI protocol with contracts to incentivize parties to adhere to the protocol. We apply the adversarial level agreements (ALAs) framework of [25] and prove lower bounds on the necessary payouts to guarantee that the best strategy of a rational party is to adhere to the protocol. We note that this approach is protocol-agnostic and thus applies to *any* Partial-APSI protocol with a similar structure, which we elaborate on below.

Our approach. When a receiver $\mathcal{C}$ initiates an intersection with a sender $\mathcal{S}$, the parties must agree on the following:

- The sender $\mathcal{S}$ specifies a minimum threshold $p \in (0, 1]$ that denotes the fraction of elements the receiver must reveal to the judge. This threshold must be met during the Authorize phase of the protocol.
- The receiver $\mathcal{C}$ can accept or reject the threshold p by taking into account the following:
  1) The loss incurred by revealing a p fraction of its elements to the judge.
  2) The utility gained by participating in the protocol.

As we will see, the payout for being caught cheating is inversely related to p. By setting a higher payout, we can tolerate a smaller fraction p while still ensuring that following the protocol would be in the receiver's best interest.

5.1. Preliminaries

Adversarial-level agreements. We augment our Partial-APSI protocol with contracts that specify payouts for when a party is caught cheating. These contracts are referred to as *adversarial-level agreements*.

Definition 5.1 ([25])**.** *Let* Π *be a cryptographic protocol with security parameter* λ *between parties* $\mathbf{P}_1, \ldots, \mathbf{P}_n$*. An* adversarial level agreement $\mathsf{ALA}_{\Pi,\kappa}(\mathbf{d})$ *consists of: (1) a specification of* Π*, defining each party's prescribed behavior and (2) damages* d_i *due from each party* $\mathbf{P}_i$ *if found deviating from* Π*, where* $\mathbf{d} = (d_1, \ldots, d_m)$ *is a vector that describes the damages for each party.*

Including adversarial level agreements (ALAs) introduces strategic interactions between rational parties. These interactions are formalized as a cryptographic inspection game (CIG).

Definition 5.2 ([25])**.** *A* cryptographic inspection game *(CIG) is a two-party game between an inspectee who wishes to deviate from a protocol and an inspector who wishes to enforce honest behavior. CIGs may have false negatives (but no false positives), and the inspector may also act dishonestly. These games are represented as trees, where the nodes correspond to the possible actions of the players and the leaves correspond to the potential outcomes.*

Any ALA-enhanced cryptographic protocol can be represented as a CIG. We aim to describe the CIG for our concrete Partial-APSI protocol and then analyze the game to find parameters that guarantee the parties will follow the protocol. Please refer to [25] for a more in-depth discussion of adversarial level agreements and inspection games.

Notation. We use the notation convetion of [25]. We denote the expected utility for a player $\mathbf{P}$ with strategy S as $\mathrm{U}_{\mathbf{P}}[S]$; this expectation is taken over the possible outcomes induced by the distribution of the other players' strategies and the game's randomness. A player's *best response* is the strategy that yields the highest expected utility given the other players' strategies. A *rational party* will act according to this best response.

Partial-APSI Structure. Our Partial-APSI protocol has the following structure, which also applies to the Partial-APSI protocol of [21] and can be generalized to future schemes. This structure will guide how we define the tree representing the Partial-APSI CIG.

- **Authorize:** The receiver and the judge first execute Authorize, during which the judge $\mathcal{J}$ randomly selects a fraction p of the receiver's element to be revealed. The receiver $\mathcal{C}$ then sends the requested elements to the judge and, if the elements are valid, the judge provides a (set of) signature(s) to $\mathcal{C}$ authorizing all of their elements.
- **Intersect:** The receiver and the sender can then execute Intersect. The receiver starts by sending $\hat{X}$ and the signature(s) (obtained during the authorization phase) to the sender $\mathcal{S}$. The sender then sends a response from which the receiver can recover the set $(\mathcal{X}_{\text{AUTH}} \cap \mathcal{X}) \cap \mathcal{Y}$, where $\mathcal{X}_{\text{AUTH}}$ is the set of items authorized by the judge.

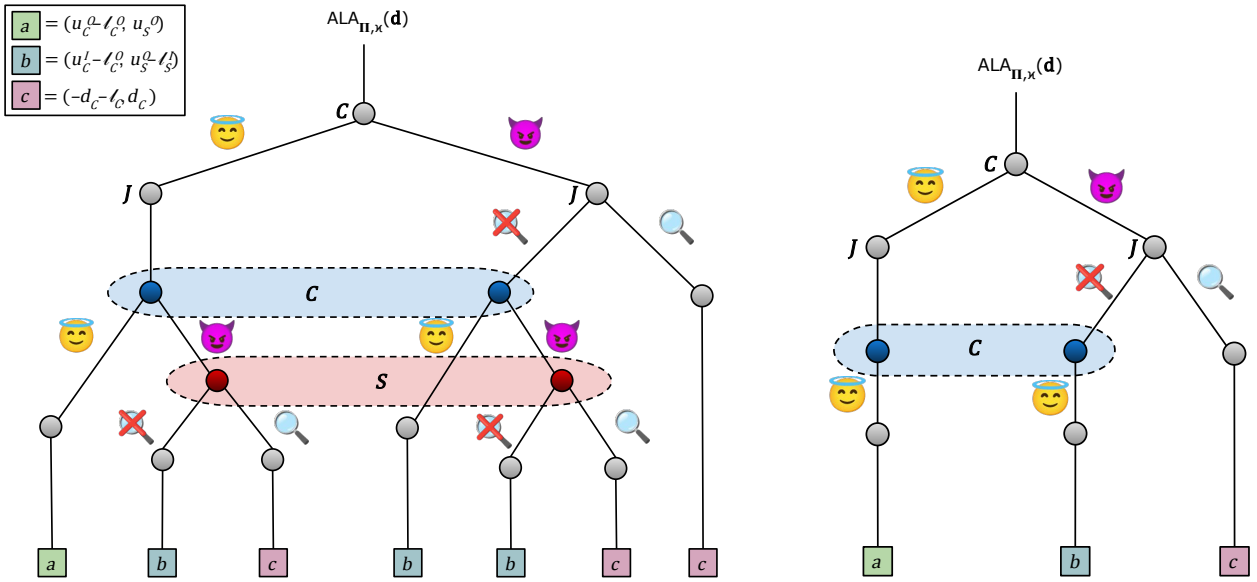
We allow the receiver to deviate from the protocol by changing their input, right before either Authorize or Intersect. Our CIG captures these two decisions.

Adversarial setting. We consider the setting of a rational receiver and honest judge/sender. In this setting, both the sender and the judge adhere to the protocol, but the receiver can deviate from the protocol by adding elements to its set to maximize its utility.

5.2. Rational Receiver and Honest Judge / Sender

In Fig. 4, we depict, as a tree, the cryptographic inspection game (CIG) for our PAPSI protocol. Each node in the tree represents a point in time when a party must make a decision, with the outgoing edges below that node corresponding to the possible actions the party can take.

For example, in Fig. 4(a), the child of the root labeled $\mathcal{C}$ represents the point during the Authorize phase when the receiver must decide whether or not to inject malicious elements into its set. The left downward edge corresponds to honest behavior, while the right edge represents the decision to inject malicious elements. The child nodes labeled with J represent the detection of malicious behavior. The dark blue children of this node correspond to the receiver (not) adding malicious elements before the Intersect phase. Next, consider the children labeled by J: the left child corresponds to the case where the receiver behaved honestly. In this case, any subset of elements sampled by the judge will be valid, so the judge approves the input with 100% probability. As such, this node has only one downward edge, representing successful approval. The right child corresponds to the case where the receiver injected malicious elements. Here, when the judge samples a p-fraction of the receiver's elements, there's a chance a malicious one is selected. The left edge represents the judge approving the input (no malicious elements detected), while the right edge corresponds to detecting a malicious element and rejecting the input set. Finally, let's examine the children of the judge nodes that correspond to the receiver. Consider, specifically, the two nodes depicted in dark blue. Once again, the receiver is



(a) The full extensive form game.

(b) The simplified game for our protocols.

Figure 4. The cryptographic inspection game for Partial-APSI between a rational receiver and an honest sender. The blue and red nodes contained in the dotted ovals form the information sets of the receiver $\mathcal{C}$ and sender $\mathcal{S}$, respectively. Here, the devil icon signifies deviation from the protocol, while the angel icon signifies adherence to the protocol. The magnifying glass denotes whether the deviation was detected.

faced with a decision: it may either act honestly (left downward edges) or inject malicious elements into the input of the Intersect phase (right downward edges).

At this point, we make an important observation: in our Partial-APSI protocol (Fig. 3), the sender will *always* detect if the receiver modifies their input set between the Authorize and Intersect phases, because the signature verification will then fail. Since we are considering a covert adversary (i.e., one that only deviates when it can avoid detection), we can conclude that a covert receiver will *never* choose to inject malicious elements at this point. As a result, we can simplify the tree on the left, resulting in the tree in Fig. 4(b).

We now turn our attention to the leaves of the CIG tree, which are labeled with payoff pairs for the receiver, $\mathcal{C}$, and the sender, $\mathcal{S}$. When the receiver behaves honestly, they learn the intersection and receive a payoff of $u_{\mathcal{C}}^{O} - \ell_{\mathcal{C}}^{O}$, where $u_{\mathcal{C}}^{O}$ is the utility of learning the intersection, and $\ell_{\mathcal{C}}^{O}$ is the loss from revealing a p-fraction of their elements. The sender, who learns nothing, receives a payoff of $u_{\mathcal{S}}^{O}$, representing the utility gained from participating in the protocol.

When the receiver injects malicious elements into their set before the Authorize phase and these are undetected, they potentially gain additional information about the sender's input beyond the intersection. In this case, the receiver earns a payoff of $u_{\mathcal{C}}^{I} - \ell_{\mathcal{C}}^{O}$, where $u_{\mathcal{C}}^{I}$ denotes the utility of learning both the intersection and any extra elements leaked from the sender's set, and $\ell_{\mathcal{C}}^{O}$ is the cost of revealing a p-fraction of their own elements. The sender, who is unaware of the deviation, learns nothing and receives a payoff of $u_{\mathcal{S}}^{O} - \ell_{\mathcal{S}}^{I}$, where $\ell_{\mathcal{S}}^{I}$ reflects the loss incurred from unintended leakage of their input.

If the receiver deviates and is caught deviating, the protocol aborts. It might be clear that the receiver thus does not learn the intersection. According to the ALA, the receiver must pay the payoff, $d_{\mathcal{C}}$, to the sender. To ensure this happens, the payoff might take the form of a deposit. Additionally, the receiver revealed the p-fraction of its elements and thus also loses $-\ell_{\mathcal{C}}^{O}$.

In Fig. 4, the blue and red nodes denote an information set for $\mathcal{C}$ and $\mathcal{S}$, respectively. An *information set* for a player is the nodes that are indistinguishable to the player. Therefore, the subtrees under those nodes are identical. See, for example, the red nodes; whether or not the receiver had malicious input in the authorization phase, the sender's point of view remains the same, and the subtrees are identical.

We now analyze the game in Fig. 4(b) to find the relationship of the above parameters, utilities, and losses that guarantee that a rational adversary is incentivised to follow the protocol. Let μ be the fraction of malicious elements that a malicious receiver would inject, and let p denote the fraction of elements revealed to the judge. If the receiver deviates and has μN malicious elements before the authorization, the probability that the judge chooses pN without selecting a malicious element is:

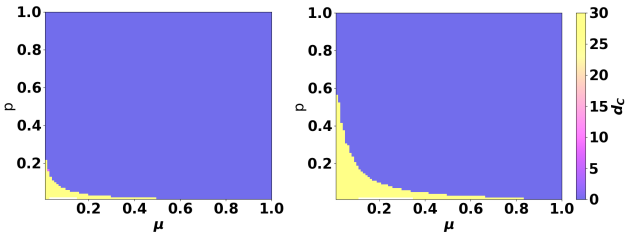
$$A = \frac{\binom{N-\mu N}{pN}}{\binom{N}{Np}} \tag{5.1}$$

Given this probability, we obtain the following lemma.

Lemma 5.3. *Let Π be a two-party Partial-APSI protocol between a rational receiver and an honest sender augmented by $\mathsf{ALA}_{\Pi,\kappa}(\mathbf{d})$ where $\mathbf{d} = (d_{\mathcal{C}}, d_{\mathcal{S}})$ specifies the damages paid by $\mathcal{C}$ and $\mathcal{S}$, respectively. If*

$$d_{\mathcal{C}} \geq \frac{-u_{\mathcal{C}}^{O} + (1-A)u_{\mathcal{C}}^{I}}{A} \tag{5.2}$$

where A is defined in (5.1), then the dominant strategy for a receiver is to follow the protocol.



(a) Shoe store. (b) Real estate.

Figure 5. We show the damages d_C needed to incentivise a rational receiver to follow the protocol. (right) Shoe store, Section 5.3.1. (left) Real estate, Section 5.3.2.

Proof. First, observe that the receiver's expected utility for not deviating is $\mathrm{U}_{\mathcal{C}}[\mathsf{NoDeviation}] = u_{\mathcal{C}}^{O} - \ell_{\mathcal{C}}^{O}$. We now compute the receiver's expected utility for deviating:

$$\begin{aligned}
&\mathrm{U}_{\mathcal{C}}[\mathsf{Deviation}] \\
&= (1-A)(u_{\mathcal{C}}^{I} - \ell_{\mathcal{C}}^{O}) + A(-d_{\mathcal{C}} - \ell_{\mathcal{C}}^{O}) \\
&= (1-A)u_{\mathcal{C}}^{I} - \ell_{\mathcal{C}}^{O} + A\ell_{\mathcal{C}}^{O} - Ad_{\mathcal{C}} - A\ell_{\mathcal{C}}^{O}) \\
&= (1-A)u_{\mathcal{C}}^{I} - \ell_{\mathcal{C}}^{O} - Ad_{\mathcal{C}} \\
&\leq (1-A)u_{\mathcal{C}}^{I} - \ell_{\mathcal{C}}^{O} - A\frac{-u_{\mathcal{C}}^{O} + (1-A)u_{\mathcal{C}}^{I}}{A} \\
&= (1-A)u_{\mathcal{C}}^{I} - \ell_{\mathcal{C}}^{O} + u_{\mathcal{C}}^{O} - (1-A)u_{\mathcal{C}}^{I} \\
&= u_{\mathcal{C}}^{O} - \ell_{\mathcal{C}}^{O}
\end{aligned}$$

where A is defined as above. The term highlighted in blue is the expected utility when the receiver deviates and the judge does not detect the deviation. The term in pink is the expected utility when the receiver deviates and the judge detects the deviation. The inequality on the third line follows from substituting inequality (5.2) Since the receiver's expected utility from deviating is upper-bounded by its expected utility from following the protocol, a rational receiver will act honestly, and the lemma follows. □

5.3. Example Scenarios

5.3.1. Shoe Store. Let us consider an e-commerce $\mathcal{C}$ that sells shoes. They want to promote a new type of shoe. $\mathcal{C}$ buys a shopping advertisement on Google ($\mathcal{S}$). The cost-per-click of this advertisement is \$0.69 [50]. The online store wishes to get the intersection between Google users who saw the ad and clients who purchased the shoes. To this end, $\mathcal{C}$ has a set $\mathcal{X}$ of $n = 1024$ IP addresses of clients and $\mathcal{S}$ a set $\mathcal{Y}$ of $m = 65536$ users.
Intersection: By learning the intersection, the store can determine if ad viewers are new to the store, have purchased additional items, etc. They can change their ad and advertising methodology accordingly, e.g., via free channels. The utility of learning 1 element in the intersection is \$0.69. If the intersection size is $0.5n$, $u_{\mathcal{C}}^{O} = \$0.69 \cdot 0.5n = \$353.28$
Cheating: All those $y \in \mathcal{Y} \backslash \{\mathcal{X} \cap \mathcal{Y}\}$ have seen the ad, but did not buy the shoes of $\mathcal{C}$. Instead, they might have bought different shoes from a competitor. Knowing this provides the store with information on which shoes to sell. We assume that one extra customer results in one extra purchase with a profit of \$20. Moreover, $\mathcal{C}$ could guess elements of $\mathcal{Y}$ by adding IP addresses of their competitors' customers. Concluding $u_{\mathcal{C}}^{I} = u_{\mathcal{C}}^{O} + \$20\mu n$.
Authorization: Finally, revealing elements for authorization might be harmful. If the judge sells this information to other online stores, they lose their customers with the \$20 profit. This results in $l_{\mathcal{C}}^{O} = \$20 \cdot pn$.

In Fig. 5(a), we can see that for $p \geq 20\%$, no damages are needed to incentivize a rational receiver (shoe store) to adhere to the protocol. For lower values of p, a damage of \$30 is preferred.

5.3.2. Real Estate. With similar reasoning, we consider the example of a website offering bankruptcy legal help. For them, the cost-per-click for a search ad is \$6.75[50], and the number of customers is likely much lower, $n = 256$. With an intersection size of $0.5n$, this results in $u_{\mathcal{C}}^{O} = 6.75 \cdot 0.5n = \$864$. Assume they earn $\$4,015$ per customer [6] with a profit of 30%, resulting in \$1204.50 per client. With a similar cheating evaluation as the online store, we obtain $u_{\mathcal{C}}^{I} = u_{\mathcal{C}}^{O} + \$858\mu$. A similar loss of revealing clients gives $l_{\mathcal{C}}^{O} = 1204.50pn$

In contrast to the shoe store (Fig. 5(a)), we can see from Fig. 5(b) that the bankruptcy lawyer does need damages for the vast majority of p. Namely, the lawyer has much more to gain by deviating. Only for $p \geq 60\%$, no damages are needed. If a lower value of p is required and the introduction of damages does not pose problems, a lower value of p can be chosen and damages of \$30 can be introduced. We conclude that the optimal damage depends on the use case and the percentage p.

6. Experimental results

To assess our Partial-APSI protocol, we implement it in Rust and evaluate it using sets of IP addresses generated with Faker [19]. First and foremost, we should choose an appropriate vector commitment scheme. In Section 3 we discussed several candidates. For the Pointproofs, we use Algorand's implementation [1]. For the Merkle tree, we use the Rust crate [58]. Algorand's Pointproofs are limited to sets of size 2^{16}. We do wish to assess our protocol for larger sets; thus, we use Pointproofs for the Verkle tree. We instantiate our protocol with the Verkle and Merkle trees. We benchmarked our protocol on an AMD Ryzen Threadripper 7970X 32-cores, 64 CPU's. Our implementation is available on https://anonymous.4open.science/r/Bandwidth_Efficient_PAPSI-0448.

6.1. Different vector commitment schemes

Here, we evaluate which vector commitment scheme suits our protocol best. We experimentally compare the use of a Merkle tree versus a Pointproof-Verkle tree.

Width of the tree. For each set size n the optimal width to reduce the runtime of our protocol is different. We thus start by determining the width k of the Verkle tree, see Fig. 8. We highlight the optimal width with a black dot. We used $m = 2^{10}$, $p = 20\%$, and the results are the average of 10 runs. Unfortunately, we could not run the pointproof protocol for widths larger than 2^{16} due to limitations in the library. In Appendix C we show that

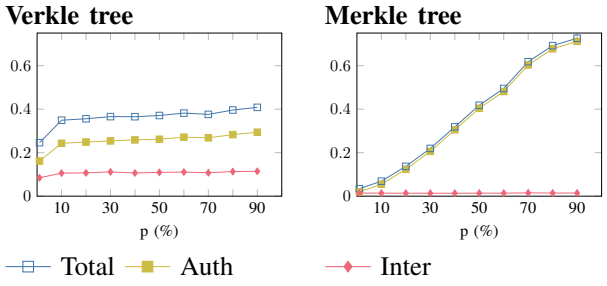


Figure 6. Runtime for different p, $m = 2^{10}$, and $n = 2^{16}$

the percentage p does not influence the width of the tree a lot. We thus choose $p = 20\%$ because this aligns with the use case of Fig. 5, no additional damages are needed to incentivize the rational store to adhere to the protocol if $p = 20\%$.

Evaluation across different p. In Fig. 6, we examine our protocol's performance across different values of p. We observe that when our protocol is instantiated with a Verkle tree, the total runtime stabilizes at around $p > 4\%$. In contrast, with a Merkle tree, this happens at around $p > 80\%$. We expect this convergence once all inner nodes appear in a proof of commitment. For $n = 2^{16}$, $k = 2^8$, and $p = 4\%$, we ask: how likely is it to sample at least one leaf per parent node? For uniform sampling, the expected value is 99%. For the Verkle tree, we can thus expect that an increase in p above 4% does not result in a huge increase in time. For the Merkle tree, with 2 children per parent, this convergence is much later.

We choose $p = 20\%$ for the remaining comparisons, as this aligns with our use case and allows us to run the Merkle tree in a reasonable amount of time.

Bandwidth. We investigate the difference in bandwidth of our protocol when using the two trees, with $m = 2^{10}$ and $p = 20\%$. Fig. 7 plots the bandwidths for various n. We conclude that the Verkle tree uses less bandwidth than the Merkle tree. This is to be expected because the proof length of the Verkle tree is $\log_k(n)$ and for the Merkle tree it is $\log_2(n)$. To illustrate, for $n = 2^{16}$, and $k = 2^8$, $\log_k(n) = 2$, whereas $\log_2(n) = 16$.

Runtime Verkle and Merkle tree. To determine the best tree for our protocol, we compare the runtime of the protocol for both trees in the Appendix; Table 3, Appendix E. In total time, the Merkle tree outperforms the Verkle tree for $n \leq 2^{16}$, but is slower for $n \geq 2^{20}$. We observe the same trend for both the Client and the Judge. For the server, the Merkle tree always performs better. For increasing n and p (Fig. 6), the protocol performs better

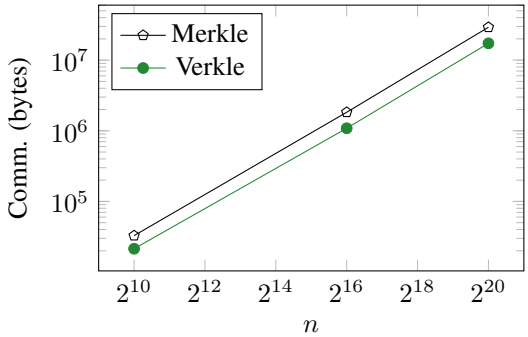


Figure 7. Plots of total bandwidth in bytes for $m = 2^{20}$, $p = 20\%$ and different n.

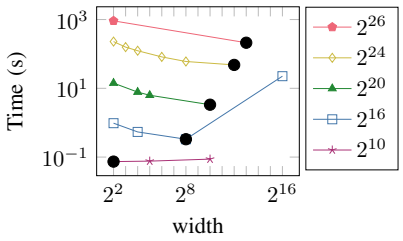


Figure 8. The total time of the protocol for different widths. Each color corresponds to a different n. $m = 2^{10}$ and $p = 20\%$. The black dots indicate the optimal width.

with the Verkle tree. Taking this into account, along with the fact that the Verkle tree requires less bandwidth, we decide to continue with the Pointproofs-Verkle tree.

6.2. Comparison with Related Work

In Table 2, we experimentally compare our protocol against Falzon and Markatou's [21] implementation [20], written in C++. Both protocols are parallelized.

Bandwidth. We observe that our protocol requires less bandwidth than Falzon and Markatou [21] for all n and m. In fact, for $n, m = 2^{20}$, we require approximately $21.2\times$ less bandwidth. There are two primary reasons for this: first, the elements of our blinded sets are smaller, around 10 bytes instead of 32. Because the elements of [21] must be defined over a field that supports bilinear pairings. Secondly, in the authorization phase of [21], the number of commitments and signatures is linear to n. In contrast, in our protocol, we send only one commitment and one signature. Thus, we achieve lower asymptotic complexity for the authorization phase.

Runtime. Regarding general runtime, we tend to outperform [21], as m increases. This is primarily because the intersection of Falzon and Markatou [21] requires computing and evaluating m OPRFs. In our protocol, the sender needs to build the verkle tree, which grows with n. Additionally, for the authorization phase, the judge in the protocol of [21] must check n EEA proofs (proving all elements are raised to the same power r, without revealing r) and compute n signatures. In our protocol, no EEA proofs are needed, and the judge signs a single token; these two improvements reduce the judge's runtime. For set sizes $n = m = 2^{20}$, and $p = 20\%$ we are about $5.9\times$ faster over a LAN network.

Cost. We compare the costs of the protocols using the c4d-standard-64 machine on Google Cloud [29], which is

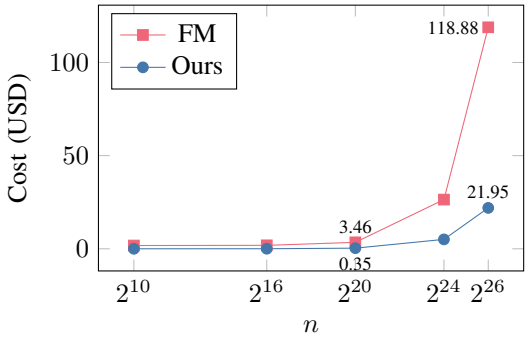


Figure 9. Costs of running the protocol 100 times for $m = 2^{20}$, $p = 20\%$ and different n.

TABLE 2. COMPARISON OF OUR PAPSI PROTOCOL WITH [21] ON AN AMD RYZEN THREADRIPPER 7970X 32-CORE CPU. WE USE DATASETS WITH IP ADDRESSES AND $p = 20$. ALL NUMBERS ARE ROUNDED TO THE NEAREST INTEGER. WE HIGHLIGHT CASES WHERE WE OUTPERFORM [21]. WE TAKE THE AVERAGE OF 10 RUNS, APART FORM THE ROW INDICATED WITH * TO AVOID ABORTION OF THE RUN.

Set sizes		Scheme	Communcation (KB)		Runtime (ms)			Total Runtime			
m	n		Auth	Inter	$\mathcal{J}$	$\mathcal{C}$	$\mathcal{S}$	LAN	1Gbps	200Mbps	50Mbps
2^{10}	2^{10}	FM [21]	146	295	1	8	6	15	575	575	575
		Verkle	9	12	21	31	20	73	73	73	73
	2^{16}	FM [21]	9507	6489	76	646	6	728	1288	1288	1288
		Verkle	557	528	35	219	95	349	349	349	349
	2^{20}	FM [21]	152082	100860	1116	9807	6	10930	11490	11490	11490
		Verkle	8907	8393	165	5722	1028	6915	6916	6917	6919
	2^{24}	FM [21]	2432504	1610810	19400	172531	12	191945	192505	192505	192505
		Verkle	142499	134222	1374	35243	10942	47558	47561	47570	47603
	2^{26}	FM [21]	9739449	6442648	79619	730126	13	809759	810319	810319	810321
		Verkle	569990	536875	5026	166448	40181	211654	211663	211699	211831
2^{16}	2^{10}	FM [21]	146	12746	1	51	481	533	1093	1093	1093
		Verkle	9	270	20	42	35	96	97	97	97
	2^{16}	FM [21]	9516	18939	77	616	486	1181	1741	1741	1741
		Verkle	557	786	41	219	104	364	364	365	365
	2^{20}	FM [21]	152255	113311	1172	9873	488	11534	12094	12094	12094
		Verkle	8907	8651	169	2232	907	3308	3309	3309	3311
	2^{24}	FM [21]	2435286	1623261	18635	171663	495	190794	191354	191354	191354
		Verkle	142499	134480	1416	35427	10972	47816	47818	47827	47860
	2^{26}	FM [21]	9737824	6455099	76906	724382	525	801815	802375	802375	802377
		Verkle	569990	537133	4796	166218	40779	211793	211802	211838	211970
2^{20}	2^{10}	FM [21]	147	202473	1	750	8332	9083	9643	9643	9643
		Verkle	9	4203	21	58	124	202	203	203	203
	2^{16}	FM [21]	9520	208666	78	1301	8361	9741	10301	10301	10301
		Verkle	557	4719	37	239	162	438	438	438	439
	2^{20}	FM [21]	152256	303038	1196	10558	8390	20145	20705	20705	20705
		Verkle	8907	12583	168	2263	979	3410	3411	3412	3414
	2^{24}	FM [21]	2436770	1812987	19454	171482	8357	199295	199855	199855	199855
		Verkle	142499	138412	1382	35356	11076	47814	47817	47826	47859
	2^{26}	FM [21]	9738788	6644826	75754	727536	8399	811690	812250	812250	812252
		Verkle	569989	541065	4832	166082	40176	211090	211099	211135	211268
2^{24}	2^{10}	FM [21]	146	3238100	1	14280	148125	162406	162966	162966	162966
		Verkle	9	67117	17	1302	1181	2500	2501	2503	2511
	2^{16}	FM [21]	9510	3244294	77	14138	147893	162109	162669	162669	162669
		Verkle	557	67633	37	1454	1254	2745	2746	2749	2757
	2^{20}	FM [21]	152136	3338665	1168	23451	147832	172452	173012	173012	173012
		Verkle	8907	75498	166	3542	2060	5769	5770	5772	5783
	2^{24}	FM [21]	2436698	4848615	19420	186636	148426	354483	355043	355043	355044
		Verkle	142499	201327	1388	37061	12130	50579	50582	50593	50635
	2^{26}	FM [21]*	9761513	9680453	78776	782378	152182	1013336	1013896	1013896	1013899
		Verkle	569981	603980	4959	169767	41720	216446	216456	216493	216634

most similar to our machine[1]. We assume the parties are in different parts of the world: the judge at the International Criminal Court in the Netherlands, the receiver in Belgium, and the sender at Google headquarters in Mountain-view California (US), with the closest machine in Oregon. The different costs of the virtual machines are; $2.96 per hour in Oregon, $3.11 per hour in the Netherlands and $3.26 per hour in Belgium [29]. The network costs are $0.02 for EU-EU communication and $0.05 for EU-US communication [28]. Fig. 9 plots the cost of running the protocol. We see that we massively reduce the costs of [21]. In fact, for $n, m = 2^{20}$, our protocol is around 10 times cheaper and for $n = 2^{26}$ around 5 times.

1. Both machines have an AMD kernel, an x86 architecture, 64 CPUs, and approximately 248 GiB of memory [30]

7. Conclusion

We present a Partial-APSI protocol with significantly improved bandwidth compared to previous work [21]. We improve on bandwidth because the field elements in our implementation require less bandwidth than those in [21]. In the authorization phase we additionally require less bandwidth for increasing n achieving lower asymptotic bandwidth. During the intersection phase our bandwidth is linear in $2n + m$ whereas the protocol of Falzon and Markatou is linear in $3m$, giving different tradeoffs. We

experimentally compare our protocol with Falzon and Markatou [21]. We concluded that for $n = m = 2^{20}$, we require approximately $21.2\times$ less bandwidth and are about $5.9\times$ faster over a LAN network. We also provide a game-theoretic analysis, demonstrating that by incorporating appropriate payouts, a rational receiver can be incentivized to adhere to the protocol.

Several interesting research directions stem from this work. First, it would be interesting to further optimize Partial-APSI protocols in terms of bandwidth and runtime, especially in the malicious setting. Secondly, there exist settings where we would like to prevent maliciously chosen inputs to PSI protocols, but no appropriate judge can be found. This leads to the question of whether we can develop such protocols without a trusted third party. Finally, our preliminary game-theoretic analysis on Partial-APSI could be extended to other MPC protocols.

References

[1] Algorand. *Pointproofs vector commitment parameter generation*. URL: https://github.com/algorand/pointproofs-paramgen/commits/master/.

[2] Yonatan Aumann and Yehuda Lindell. "Security against covert adversaries: efficient protocols for realistic adversaries". In: *Proceedings of the 4th Conference on Theory of Cryptography*. TCC'07. Amsterdam, The Netherlands: Springer-Verlag, 2007, pp. 137–156. ISBN: 9783540709350.

[3] Pablo Daniel Azar and Silvio Micali. "Rational proofs". In: *Proceedings of the Forty-Fourth Annual ACM Symposium on Theory of Computing*. STOC '12. New York, New York, USA: Association for Computing Machinery, 2012, pp. 1017–1028. ISBN: 9781450312455. DOI: 10.1145/2213977.2214069. URL: https://doi.org/10.1145/2213977.2214069.

[4] Pierre Baldi et al. "Countering GATTACA: efficient and secure testing of fully-sequenced human genomes". In: *Proceedings of the 18th ACM Conference on Computer and Communications Security*. CCS '11. Chicago, Illinois, USA: Association for Computing Machinery, 2011, pp. 691–702. ISBN: 9781450309486. DOI: 10.1145/2046707.2046785. URL: https://doi.org/10.1145/2046707.2046785.

[5] Dan Boneh. "The Decision Diffie-Hellman Problem". In: *Algorithmic Number Theory, Third International Symposium, ANTS-III, Portland, Oregon, USA, June 21-25, 1998, Proceedings*. Ed. by Joe Buhler. Vol. 1423. Lecture Notes in Computer Science. Portland, Oregon, USA: Springer, 1998, pp. 48–63. DOI: 10.1007/BFB0054851. URL: https://doi.org/10.1007/BFb0054851.

[6] Catherine Brock. What is the average lawyer retainer fee? Bankruptcy. URL: https://www.lawpay.com/about/blog/lawyer-hourly-rate-by-state/.

[7] Jan Camenisch and Gregory M. Zaverucha. "Private Intersection of Certified Sets". In: *Financial Cryptography and Data Security*. Ed. by Roger Dingledine and Philippe Golle. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009, pp. 108–127. ISBN: 978-3-642-03549-4.

[8] Dario Catalano and Dario Fiore. "Vector Commitments and Their Applications". In: *Public-Key Cryptography - PKC 2013 - 16th International Conference on Practice and Theory in Public-Key Cryptography, Nara, Japan, February 26 - March 1, 2013. Proceedings*. Ed. by Kaoru Kurosawa and Goichiro Hanaoka. Vol. 7778. Lecture Notes in Computer Science. Nara, Japan: Springer, 2013, pp. 55–72. DOI: 10.1007/978-3-642-36362-7_5. URL: https://doi.org/10.1007/978-3-642-36362-7_5.

[9] Melissa Chase and Peihan Miao. "Private Set Intersection in the Internet Setting from Lightweight Oblivious PRF". In: *Advances in Cryptology – CRYPTO 2020: 40th Annual International Cryptology Conference, CRYPTO 2020, Santa Barbara, CA, USA, August 17–21, 2020, Proceedings, Part III*. Santa Barbara, CA, USA: Springer-Verlag, 2020, pp. 34–63. ISBN: 978-3-030-56876-4. DOI: 10.1007/978-3-030-56877-1_2. URL: https://doi.org/10.1007/978-3-030-56877-1_2.

[10] Hao Chen, Kim Laine, and Peter Rindal. "Fast Private Set Intersection from Homomorphic Encryption". In: *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security, CCS 2017, Dallas, TX, USA, October 30 - November 03, 2017*. Ed. by Bhavani Thuraisingham et al. ACM, 2017, pp. 1243–1255. DOI: 10.1145/3133956.3134061. URL: https://doi.org/10.1145/3133956.3134061.

[11] CleanPegasus. *kzg-commitment*. URL: https://crates.io/crates/kzg-commitment.

[12] Paolo D'Arco et al. "Size-Hiding in Private Set Intersection: Existential Results and Constructions". In: *Progress in Cryptology - AFRICACRYPT 2012*. Ed. by Aikaterini Mitrokotsa and Serge Vaudenay. Berlin, Heidelberg: Springer Berlin Heidelberg, 2012, pp. 378–394. ISBN: 978-3-642-31410-0. DOI: 10.1007/978-3-642-31410-0_23.

[13] Emiliano De Cristofaro, Jihye Kim, and Gene Tsudik. "Linear-Complexity Private Set Intersection Protocols Secure in Malicious Model". In: *Advances in Cryptology - ASIACRYPT 2010*. Ed. by Masayuki Abe. Berlin, Heidelberg: Springer Berlin Heidelberg, 2010, pp. 213–231. ISBN: 978-3-642-17373-8. DOI: 10.1007/978-3-642-17373-8_13.

[14] Emiliano De Cristofaro and Gene Tsudik. "Practical Private Set Intersection Protocols with Linear Complexity". In: *Financial Cryptography and Data Security*. Ed. by Radu Sion. Berlin, Heidelberg: Springer Berlin Heidelberg, 2010, pp. 143–159. ISBN: 978-3-642-14577-3. DOI: 10.1007/978-3-642-14577-3_13.

[15] Emiliano De Cristofaro et al. "Privacy-Preserving Policy-Based Information Transfer". In: *Privacy Enhancing Technologies*. Ed. by Ian Goldberg and Mikhail J. Atallah. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009, pp. 164–184. ISBN: 978-3-642-03168-7.

[16] Sumit Kumar Debnath and Ratna Dutta. "Secure and Efficient Private Set Intersection Cardinality Using Bloom Filter". In: *Information Security*. Ed. by Javier Lopez and Chris J. Mitchell. Cham:

Springer International Publishing, 2015, pp. 209–226. ISBN: 978-3-319-23318-5. DOI: 10.1007/978-3-319-23318-5_12.

[17] California Privacy Protection Agency Enforcement Division. *Applying Data Minimzation to Consumer Requests*. https : / / cppa . ca . gov / pdf / enfadvisory202401 . pdf. Visited 8 July 2025. Jan. 2024.

[18] Sky Faber, Ronald Petrlic, and Gene Tsudik. "UnLinked: Private Proximity-based Off-line OSN Interaction". In: *Proceedings of the 14th ACM Workshop on Privacy in the Electronic Society, WPES 2015, Denver, Colorado, USA, October 12, 2015*. Ed. by Indrajit Ray, Nicholas Hopper, and Rob Jansen. Denver, Colorado, USA: ACM, 2015, pp. 121–131. DOI: 10 . 1145 / 2808138 . 2808149. URL: https://doi.org/10.1145/2808138.2808149.

[19] *Faker*. URL: https://faker.readthedocs.io/en/master/providers/faker.providers.internet.html.

[20] Francesca Falzon and Evangelia Anna Markatou. *Re-visiting Authorized Private Set Intersection: A New Privacy-Preserving Variant and Two Protocols*. https : / / github . com / markatou / Partial - APSI. [Online; accessed 27-May-2025]. 2025.

[21] Francesca Falzon and Evangelia Anna Markatou. "Re-visiting Authorized Private Set Intersection: A New Privacy-Preserving Variant and Two Protocols". In: *Proc. Priv. Enhancing Technol.* 2025.1 (2025), pp. 792–807. DOI: 10 . 56553 / POPETS - 2025 - 0041. URL: https://doi.org/10.56553/popets-2025-0041.

[22] Francesca Falzon and Tianxin Tang. *Learning from Functionality Outputs: Private Join and Compute in the Real World*. 2025. URL: https://www.usenix.org / system / files / conference / usenixsecurity25 / sec25cycle1-prepub-994-falzon.pdf.

[23] Dankrad Feist. *PCS multiproofs using random evaluation*. June 2021. URL: https : / / dankradfeist . de / ethereum/2021/06/18/pcs-multiproofs.html.

[24] *General Data Protection Regulation (GDPR)*. https://gdpr-info.eu/. Visited 8 July 2025. 2016.

[25] Marilyn George and Seny Kamara. "Adversarial Level Agreements for Two-Party Protocols". In: *ASIA CCS '22: ACM Asia Conference on Computer and Communications Security, Nagasaki, Japan, 30 May 2022 - 3 June 2022*. Ed. by Yuji Suga et al. Nagasaki, Japan: ACM, 2022, pp. 816–830. DOI: 10.1145/3488932.3517385. URL: https://doi.org/10.1145/3488932.3517385.

[26] Bishakh Chandra Ghosh et al. "Private certifier intersection". In: *Cryptology ePrint Archive* (2022).

[27] Aarushi Goel et al. *PICS: Private Intersection over Committed (and reusable) Sets*. Cryptology ePrint Archive, Paper 2025/1071. 2025. URL: https : / / eprint.iacr.org/2025/1071.

[28] Google. *All networking pricing*. URL: https://cloud.google.com/vpc/network-pricing.

[29] Google. *Compute Engine pricing*. URL: https : / / cloud.google.com/compute/all-pricing.

[30] Google. *Machine families resource and comparison guide*. URL: https : / / cloud . google . com / compute / docs/machine-resource#machine_type_comparison.

[31] Sergey Gorbunov et al. "Pointproofs: Aggregating Proofs for Multiple Vector Commitments". In: *CCS '20: 2020 ACM SIGSAC Conference on Computer and Communications Security, Virtual Event, USA, November 9-13, 2020*. Ed. by Jay Ligatti et al. Virtual Event, USA: ACM, 2020, pp. 2007–2023. DOI: 10.1145/3372297.3417244. URL: https://doi.org/10.1145/3372297.3417244.

[32] S. Dov Gordon and Jonathan Katz. "Rational secret sharing, revisited". In: *Proceedings of the 5th International Conference on Security and Cryptography for Networks*. SCN'06. Maiori, Italy: Springer-Verlag, 2006, pp. 229–241. ISBN: 3540380809. DOI: 10.1007/11832072_16. URL: https://doi.org/10.1007/11832072_16.

[33] Xiaojie Guo et al. "Birds of a Feather Flock Together: How Set Bias Helps to Deanonymize You via Revealed Intersection Sizes". In: *31st USENIX Security Symposium (USENIX Security 22)*. Boston, MA: USENIX Association, Aug. 2022, pp. 1487–1504. ISBN: 978-1-939133-31-1. URL: https : / / www . usenix . org / conference / usenixsecurity22 / presentation/guo.

[34] Joseph Halpern and Vanessa Teague. "Rational secret sharing and multiparty computation: extended abstract". In: *Proceedings of the Thirty-Sixth Annual ACM Symposium on Theory of Computing*. STOC '04. Chicago, IL, USA: Association for Computing Machinery, 2004, pp. 623–632. ISBN: 1581138520. DOI: 10 . 1145 / 1007352 . 1007447. URL: https://doi.org/10.1145/1007352.1007447.

[35] Laura Hetz, Thomas Schneider, and Christian Weinert. "Scaling Mobile Private Contact Discovery to Billions of Users". In: *Computer Security - ESORICS 2023 - 28th European Symposium on Research in Computer Security, The Hague, The Netherlands, September 25-29, 2023, Proceedings, Part I*. Ed. by Gene Tsudik et al. Vol. 14344. Lecture Notes in Computer Science. Springer, 2023, pp. 455–476. DOI: 10 . 1007 / 978 - 3 - 031 - 50594 - 2_23. URL: https://doi.org/10.1007/978-3-031-50594-2_23.

[36] Bernardo A. Huberman, Matt Franklin, and Tad Hogg. "Enhancing privacy and trust in electronic communities". In: *Proceedings of the 1st ACM Conference on Electronic Commerce*. EC '99. Denver, Colorado, USA: Association for Computing Machinery, 1999, pp. 78–86. ISBN: 1581131763. DOI: 10.1145/336992.337012. URL: https://doi.org/10.1145/336992.337012.

[37] Mihaela Ion et al. "On Deploying Secure Computing: Private Intersection-Sum-with-Cardinality". In: *IEEE European Symposium on Security and Privacy, EuroS&P 2020, Genoa, Italy, September 7-11, 2020*. IEEE, 2020, pp. 370–389. DOI: 10 . 1109 / EUROSP48549 . 2020 . 00031. URL: https : // doi.org/10.1109/EuroSP48549.2020.00031.

[38] Mihaela Ion et al. "Private Intersection-Sum Protocol with Applications to Attributing Aggregate Ad Conversions". In: *IACR Cryptol. ePrint Arch.* (2017), p. 738. URL: http://eprint.iacr.org/2017/738.

[39] Bo Jiang, Jian Du, and Qiang Yan. "AnonPSI: An Anonymity Assessment Framework for PSI". In:

31st Annual Network and Distributed System Security Symposium, NDSS 2024, San Diego, California, USA, February 26 - March 1, 2024. San Diego, California, USA: The Internet Society, 2024. URL: https://www.ndss-symposium.org/ndss-paper/anonpsi-an-anonymity-assessment-framework-for-psi/.

[40] Daniel Kales et al. "Mobile Private Contact Discovery at Scale". In: *28th USENIX Security Symposium, USENIX Security 2019, Santa Clara, CA, USA, August 14-16, 2019*. Ed. by Nadia Heninger and Patrick Traynor. USENIX Association, 2019, pp. 1447–1464. URL: https://www.usenix.org/conference/usenixsecurity19/presentation/kales.

[41] Aniket Kate, Gregory M. Zaverucha, and Ian Goldberg. "Constant-Size Commitments to Polynomials and Their Applications". In: *Advances in Cryptology - ASIACRYPT 2010*. Ed. by Masayuki Abe. Berlin, Heidelberg: Springer Berlin Heidelberg, 2010, pp. 177–194. ISBN: 978-3-642-17373-8. DOI: https://doi.org/10.1007/978-3-642-17373-8_11.

[42] Jonathan Katz. "Bridging game theory and cryptography: recent results and future directions". In: *Proceedings of the 5th Conference on Theory of Cryptography*. TCC'08. New York, USA: Springer-Verlag, 2008, pp. 251–272. ISBN: 354078523X.

[43] Florian Kerschbaum. "Outsourced private set intersection using homomorphic encryption". In: *7th ACM Symposium on Information, Compuer and Communications Security, ASIACCS '12, Seoul, Korea, May 2-4, 2012*. Ed. by Heung Youl Youm and Yoojae Won. Seoul, Korea: ACM, 2012, pp. 85–86. DOI: 10.1145/2414456.2414506. URL: https://doi.org/10.1145/2414456.2414506.

[44] John Kuszmaul. *Verkle Trees*. 2019. URL: https://api.semanticscholar.org/CorpusID:218475793.

[45] Tancrède Lepoint et al. "Private Join and Compute from PIR with Default". In: *Advances in Cryptology - ASIACRYPT 2021 - 27th International Conference on the Theory and Application of Cryptology and Information Security, Singapore, December 6-10, 2021, Proceedings, Part II*. Ed. by Mehdi Tibouchi and Huaxiong Wang. Vol. 13091. Lecture Notes in Computer Science. Springer, 2021, pp. 605–634. DOI: 10.1007/978-3-030-92075-3_21. URL: https://doi.org/10.1007/978-3-030-92075-3_21.

[46] Ming Li et al. "Privacy-Preserving Distributed Profile Matching in Proximity-Based Mobile Social Networks". In: *IEEE Transactions on Wireless Communications* 12.5 (2013), pp. 2024–2033. DOI: 10.1109/TWC.2013.032513.120149.

[47] Benoit Libert and Moti Yung. "Concise Mercurial Vector Commitments and Independent Zero-Knowledge Sets with Short Proofs". In: *Theory of Cryptography, 7th Theory of Cryptography Conference, TCC 2010, Zurich, Switzerland, February 9-11, 2010. Proceedings*. Ed. by Daniele Micciancio. Vol. 5978. Lecture Notes in Computer Science. Zurich, Switzerland: Springer, 2010, pp. 499–517. DOI: 10.1007/978-3-642-11799-2_30. URL: https://doi.org/10.1007/978-3-642-11799-2_30.

[48] Catherine Meadows. "A More Efficient Cryptographic Matchmaking Protocol for Use in the Absence of a Continuously Available Third Party". In: *Proceedings of the 1986 IEEE Symposium on Security and Privacy, Oakland, California, USA, April 7-9, 1986*. Oakland, California, USA: IEEE Computer Society, 1986, pp. 134–137. DOI: 10.1109/SP.1986.10022. URL: https://doi.org/10.1109/SP.1986.10022.

[49] Ralph C. Merkle. "A Certified Digital Signature". In: *Advances in Cryptology - CRYPTO '89, 9th Annual International Cryptology Conference, Santa Barbara, California, USA, August 20-24, 1989, Proceedings*. Ed. by Gilles Brassard. Vol. 435. Lecture Notes in Computer Science. Santa Barbara, California, USA: Springer, 1989, pp. 218–238. DOI: 10.1007/0-387-34805-0_21. URL: https://doi.org/10.1007/0-387-34805-0_21.

[50] Dennis Moons. *27 Google Ads Benchmarks (2025)*. Average Google Shopping CPC - Clothing & Apparel and Average Google Search Ads Cost Per Click - Legal. URL: https://www.storegrowers.com/google-ads-benchmarks/.

[51] Shishir Nagaraja et al. "BotGrep: finding P2P bots with structured graph analysis". In: *Proceedings of the 19th USENIX Conference on Security*. USENIX Security'10. Washington, DC: USENIX Association, 2010, p. 7.

[52] Marcin Nagy et al. "Do I know you?: efficient and privacy-preserving common friend-finder protocols and applications". In: *Annual Computer Security Applications Conference, ACSAC '13, New Orleans, LA, USA, December 9-13, 2013*. Ed. by Charles N. Payne Jr. Orleans, LA, USA: ACM, 2013, pp. 159–168. DOI: 10.1145/2523649.2523668. URL: https://doi.org/10.1145/2523649.2523668.

[53] Benny Pinkas et al. "Efficient Circuit-Based PSI via Cuckoo Hashing". In: *Advances in Cryptology - EUROCRYPT 2018 - 37th Annual International Conference on the Theory and Applications of Cryptographic Techniques, Tel Aviv, Israel, April 29 - May 3, 2018 Proceedings, Part III*. Ed. by Jesper Buus Nielsen and Vincent Rijmen. Vol. 10822. Lecture Notes in Computer Science. Springer, 2018, pp. 125–157. DOI: 10.1007/978-3-319-78372-7_5. URL: https://doi.org/10.1007/978-3-319-78372-7_5.

[54] Benny Pinkas et al. "SpOT-Light: Lightweight Private Set Intersection from Sparse OT Extension". In: *Advances in Cryptology – CRYPTO 2019: 39th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 18–22, 2019, Proceedings, Part III*. Santa Barbara, CA, USA: Springer-Verlag, 2019, pp. 401–431. ISBN: 978-3-030-26953-1. DOI: 10.1007/978-3-030-26954-8_13. URL: https://doi.org/10.1007/978-3-030-26954-8_13.

[55] Liyan Shen et al. "Efficient and Private Set Intersection of Human Genomes". In: *IEEE International Conference on Bioinformatics and Biomedicine, BIBM 2018, Madrid, Spain, December 3-6, 2018*. Ed. by Huiru Jane Zheng et al. New York: IEEE Computer Society, 2018, pp. 761–764. DOI: 10.1109/BIBM.2018.8621291. URL: https://doi.

ieeecomputersociety . org / 10 . 1109 / BIBM . 2018 . 8621291.

[56] Emil Stefanov, Elaine Shi, and Dawn Song. "Policy-Enhanced Private Set Intersection: Sharing Information While Enforcing Privacy Policies". In: *Public Key Cryptography – PKC 2012*. Ed. by Marc Fischlin, Johannes Buchmann, and Mark Manulis. Berlin, Heidelberg: Springer Berlin Heidelberg, 2012, pp. 413–430. ISBN: 978-3-642-30057-8. DOI: 10.1007/978-3-642-30057-8_25.

[57] Yunqing Sun et al. "Actively Secure Private Set Intersection in the Client-Server Setting". In: *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*. CCS '24. Salt Lake City, UT, USA: Association for Computing Machinery, 2024, pp. 1478–1492. ISBN: 9798400706363. DOI: 10.1145/3658644.3690349. URL: https://doi.org/10.1145/3658644.3690349.

[58] Anton Suprunchuk. *rs_merkle*. URL: https://crates.io/crates/rs_merkle.

[59] Alin Tomescu et al. "Aggregatable Subvector Commitments for Stateless Cryptocurrencies". In: *Security and Cryptography for Networks - 12th International Conference, SCN 2020, Amalfi, Italy, September 14-16, 2020, Proceedings*. Ed. by Clemente Galdi and Vladimir Kolesnikov. Vol. 12238. Lecture Notes in Computer Science. Amalfi, Italy: Springer, 2020, pp. 45–64. DOI: 10.1007/978-3-030-57990-6_3. URL: https://doi.org/10.1007/978-3-030-57990-6_3.

[60] Jelle Vos et al. *On the Insecurity of Bloom Filter-Based Private Set Intersections*. Cryptology ePrint Archive, Paper 2024/1901. 2024. URL: https://eprint.iacr.org/2024/1901.

[61] Connie M Westhoff. "Blood group genotyping". In: *Blood, The Journal of the American Society of Hematology* 133.17 (2019), pp. 1814–1820.

[62] Yunhao Yang et al. "A blind signature-based Authorization Scheme for Enhancing the Privacy of Cloud-Assisted Private Set Intersection". In: *Computer Networks* (2025), p. 111844. ISSN: 1389-1286. DOI: https://doi.org/10.1016/j.comnet.2025.111844. URL: https://www.sciencedirect.com/science/article/pii/S1389128625008102.

Appendix A.

In the appendix we justify the choice of using Pointproofs for the Verkle tree and provide some details on how our protocol scales better with a Verkle than a Merkle tree. We provide in Appendix B additional computational assumptions needed for the Pointproofs. We also add computational assumptions needed for the so called KZG vector commitment scheme. In Appendix C we evaluate the best width of the verkle tree with both KZG and Pointproofs. In Appendix D we explain why we chose the Pointproofs as the better commitment scheme for the Verkle tree. The last section of the appendix, Appendix E compares and evaluates the differences between the Merkle and Verkle tree.

Appendix B. Computational assumptions

For completeness, we define the computational assumptions needed for the KZG and pointproof commitment scheme in this section.

Definition B.1. Discrete Logarithm problem *(DL): For a generator g of group $\mathbb{G}$ of prime order p and $a \in \mathbb{Z}_p$ it holds that for all probabilistic polynomial time algorithms $\mathcal{A}$,*

$$\Pr[\mathcal{A}(g, g^a) = a] \leq \mathsf{negl}(\lambda),$$

for security parameter $\mathsf{negl}(\lambda)$.

A variant of the DDH assumption is the t-Strong Diffie-Hellman (t-SDH) assumption [41].

Definition B.2. t-Strong Diffie-Hellman (t-SDH) assumption*: Given a group $\mathbb{G}$ of prime order q and a tuple $\left\langle g, g^{\alpha}, g^{\alpha^2}, ..., g^{\alpha^t} \right\rangle$ for $g \in \mathbb{G}$, it holds that for all probabilistic polynomial time algorithms $\mathcal{A}$,*

$$\Pr\left[\mathcal{A}(g, g^{\alpha}, g^{\alpha^2}, ..., g^{\alpha^t}) = \left\langle c, g^{\frac{1}{\alpha+c}} \right\rangle\right] \leq \mathsf{negl}(\lambda),$$

for any value $c \in \mathbb{Z}_{\shortparallel} \setminus \{-\alpha\}$ and security parameter $\mathsf{negl}(\lambda)$.

A variant of the t-SDH assumption is the t-Bilinear Strong Diffie-Hellman (t-BSDH) assumption [41].

Definition B.3. t-Bilinear Strong Diffie-Hellman (t-BSDH) assumption *Consider a similar setting as for the t-SDH assumption with an additional bilinear pairing e, it holds that for all probabilistic polynomial time algorithms $\mathcal{A}$,*

$$\Pr\left[\mathcal{A}(g, g^{\alpha}, g^{\alpha^2}, ..., g^{\alpha^t}) = \left\langle c, e(g,g)^{\frac{1}{\alpha+c}} \right\rangle\right] \leq \mathsf{negl}(\lambda),$$

for any value $c \in \mathbb{Z}_{\shortparallel} \setminus \{-\alpha\}$ and security parameter $\mathsf{negl}(\lambda)$.

A variant of the t-BSDH assumption is the t-weak bilinear Diffie-Hellman assumption [31]

Definition B.4. t-weak Bilinear Strong Diffie-Hellman (t-wBSDH) assumption *Consider a similar setting as for the t-BSDH assumption, it holds that for all probabilistic polynomial time algorithms $\mathcal{A}$,*

$$\Pr\left[\mathcal{A}(g, g^{\alpha}, ..., g^{\alpha^t}, g^{\alpha^{t+1}}, ..., g^{\alpha^n}) = \left\langle c, e(g,g)^{\alpha+1} \right\rangle\right] \leq \mathsf{negl}(\lambda),$$

for security parameter $\mathsf{negl}(\lambda)$.

Definition B.5. The Algebraic Group Model (AGM) *Given a group $\mathbb{G}$ and elements $x_1, ..., x_n \in \mathbb{G}$ for all probabilistic polynomial time algorithms $\mathcal{A}$,*

$$\Pr[Z = \prod_{i=1}^{n} x_i^{z_i} \mid z_1, .., z_n, Z \leftarrow \mathcal{A}] \leq \mathsf{negl}(\lambda),$$

for security parameter $\mathsf{negl}(\lambda)$.

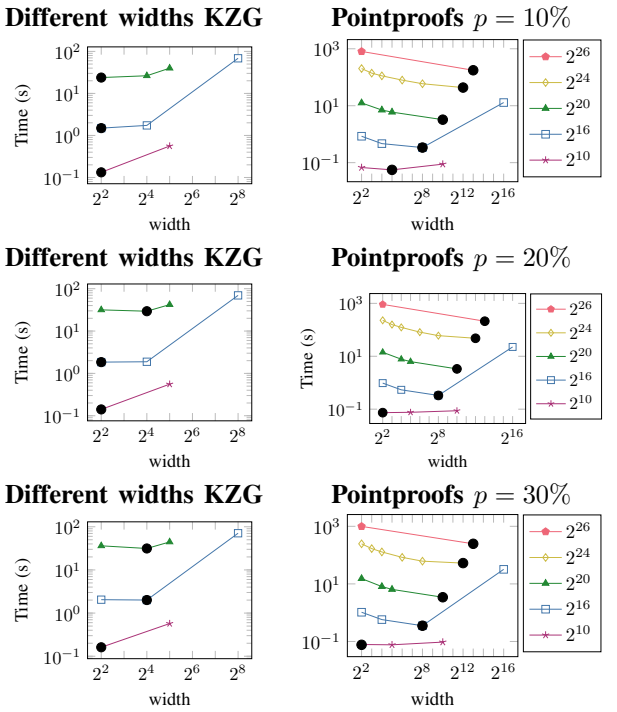


Figure 10. The total time of the protocol for $m = 2^{10}$. Each color corresponds to a different n. The black dots indicate the optimal width.

Appendix C. Widths of the tree for different p

In the paper, we determined the optimal width of the tree by evaluating the runtime of our protocol for $m = 2^{20}$, $p = 20\%$, and various values of n. It might be clear that the value of m is irrelevant for the width of the tree, but p might influence the optimal width. To assess this influence, Fig. 10 plots the runtime of the protocol for different p and widths. We conclude that width depends somewhat on p, but the dependence is not significant.

Appendix D. Why not KZG based Verkle tree?

We begin this section with an explanation of the KZG commitment scheme and then evaluate its performance.

D.1. KZG vector commitment

Kate et al. [41] proposed a commitment scheme based on polynomials, which can be adapted into a vector commitment scheme as Feist [23] shows. On a high level, the scheme takes several values m_i for $i \leq n$ and interpolates a polynomial for the set of points $\{(x_i, m_i)\}_{i \leq n}$. The commitment is the polynomial evaluated at a secret value. Fortunately, this commitment is concise.

The notion of polynomial binding is introduced to guarantee the security of a polynomial commitment scheme. This notion captures that it is unlikely to find two polynomials with the same commitment. It can be proven that the KZG commitment scheme is polynomial binding, hiding, and binding if the discrete logarithm and the t-(bilinear) strong Diffie-Hellman assumptions hold, for a definition of the assumptions see Appendix B. For

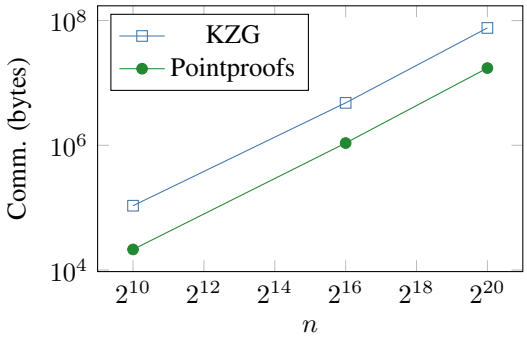


Figure 11. Plots of total bandwidth in bytes for $m = 2^{20}$, $p = 20\%$ and different n.

polynomial binding and hiding, we refer the reader to [41], for binding, we refer the reader to [59]. This vector commitment scheme supports batch openings and is concise.

D.2. Implementation

For the KZG commitments, we used the Rust crates [11]. We instantiate our protocol with a KZG Verkle tree and evaluate the optimal branching factor of the tree.

From Fig. 8, we see that the optimal width (announced with a black dot) is different for both the KZG and the Pointproofs commitment scheme. We highlight that the KZG proofs are about a factor of 10 slower than the Pointproofs.

In Fig. 11, we see that our protocol with the Pointproof-Verkle tree uses less bandwidth than the KZG-Verkle tree. This is mainly because the KZG library did not allow for serialization of the elements.

We concluded that a pointproofs-based verkle tree outperforms a KZG-verkle tree in both bandwidth and runtime. We therefore proceed with the Pointproofs-verkle tree.

Appendix E. Comparision Merkle and Verkle tree

In this section, we compare the runtime and bandwidth of our protocol using a Verkle and Merkle tree. The results are visible in Table 3. We conclude that for small values of n, ie, $n \leq 2^{16}$, the Merkle tree outperforms the Verkle tree. However, we expect from Fig. 6 that for increasing p the Verkle tree would be faster. For increasing n and $p = 20\%$, the table shows that the Verkle tree is significantly faster than the Merkle tree.

We aim to provide a Partial-APSI protocol that has reduced bandwidth compared to previous works, and we want the protocol to scale well with increasing n and p. We therefore chose the Verkle tree as most suitable for our goal. However, another setting might require the protocol to be fast on the server side; in this case, we suggest using the Merkle tree.

TABLE 3. COMPARISON OF OUR PAPSI PROTOCOL WITH A MERKLE AND VERKLE TREE ON AN AMD RYZEN THREADRIPPER 7970X 32-CORE CPU USING DATASETS OF IP ADDRESSES. THE PERCENTAGE $p = 20$, ALL NUMBERS ARE ROUNDED TO THE NEAREST INTEGER. WE HIGHLIGHT CASES WHERE THE VERKLE TREE OUTPERFORMS THE MERKLE TREE IN BLUE. WE TAKE THE AVERAGE OF 2 RUNS, SLIGHT DIFFERENCES CAN THEREFORE OCCUR WITH TABLE 2.

Set sizes		Scheme	Communcation (KB)		Runtime (ms)			Total Runtime			
m	n		Auth	Inter	$\mathcal{J}$	$\mathcal{C}$	$\mathcal{S}$	LAN	1Gbps	200Mbps	50Mbps
2^{10}	2^{10}	**Merkle**	21	12	0	0	0	1	1	1	1
		Verkle	9	12	13	21	15	49	49	49	49
	2^{16}	**Merkle**	1303	528	54	75	11	139	140	140	140
		Verkle	557	528	54	261	110	425	426	426	426
	2^{20}	**Merkle**	20817	8393	12031	14577	188	26796	26796	26797	26801
		Verkle	8907	8393	170	2 283	908	3361	3361	3362	3364
	2^{24}	**Merkle**	332954	134222	3265045	3281925	3205	6550174	6550178	6550193	6550249
		Verkle	142498	134222	1493	35616	10890	47999	48001	48010	48044
2^{16}	2^{10}	**Merkle**	21	270	0	2	5	7	7	7	7
		Verkle	9	270	20	28	28	75	76	76	76
	2^{16}	**Merkle**	1300	786	53	66	15	134	134	134	134
		Verkle	557	786	37	241	95	373	373	373	373
	2^{20}	**Merkle**	20810	8651	11915	12098	189	24203	24203	24204	24208
		Verkle	8907	8651	166	2260	891	3317	3318	3318	3320
	2^{24}	**Merkle**	332995	134480	3028496	3043304	3309	6075108	6075112	6075127	6075183
		Verkle	142498	134480	1418	35541	10889	47848	47851	47860	47893
2^{20}	2^{10}	**Merkle**	20	4203	0	30	70	100	100	100	101
		Verkle	9	4203	14	57	93	164	165	165	165
	2^{16}	**Merkle**	1299	4719	53	92	82	227	228	228	229
		Verkle	557	4719	37	270	168	475	476	476	477
	2^{20}	**Merkle**	20811	12583	11917	12168	256	24341	24342	24343	24347
		Verkle	8907	12583	167	2294	969	3431	3431	3432	3435
	2^{24}	**Merkle**	332967	138412	3327238	3035548	3333	6366119	6366123	6366138	6366194
		Verkle	142499	138412	1439	35665	11116	48219	48222	48231	48265
2^{24}	2^{10}	**Merkle**	20	67117	0	1205	1182	2388	2389	2391	2399
		Verkle	9	67117	14	1423	1248	2685	2685	2688	2696
	2^{16}	**Merkle**	1302	67633	53	1279	1158	2489	2490	2492	2501
		Verkle	557	67633	36	1629	1308	2973	2974	2977	2985
	2^{20}	**Merkle**	20806	75498	11904	13340	1319	26562	26563	26566	26578
		Verkle	8907	75498	166	3673	2090	5928	5929	5932	5942
	2^{24}	**Merkle**	332959	201327	3315120	3068674	4529	6388323	6388328	6388345	6388409
		Verkle	142498	201327	1486	37481	12098	51064	51067	51078	51120